

Setup Guide for Splunk Enterprise and Splunk Universal Forwarder (SIEM tools)

This document will contain the **full technical setup**.

Structure it like this:

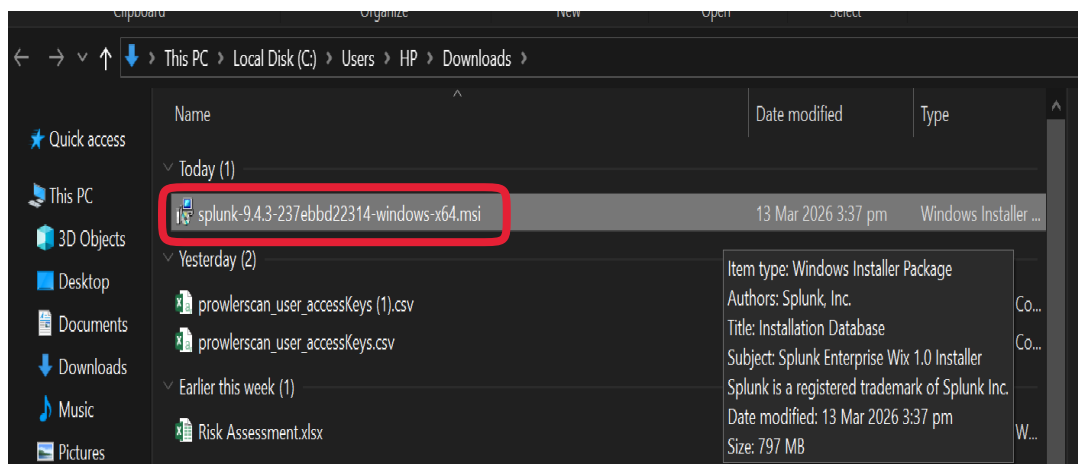
1. Lab Environment

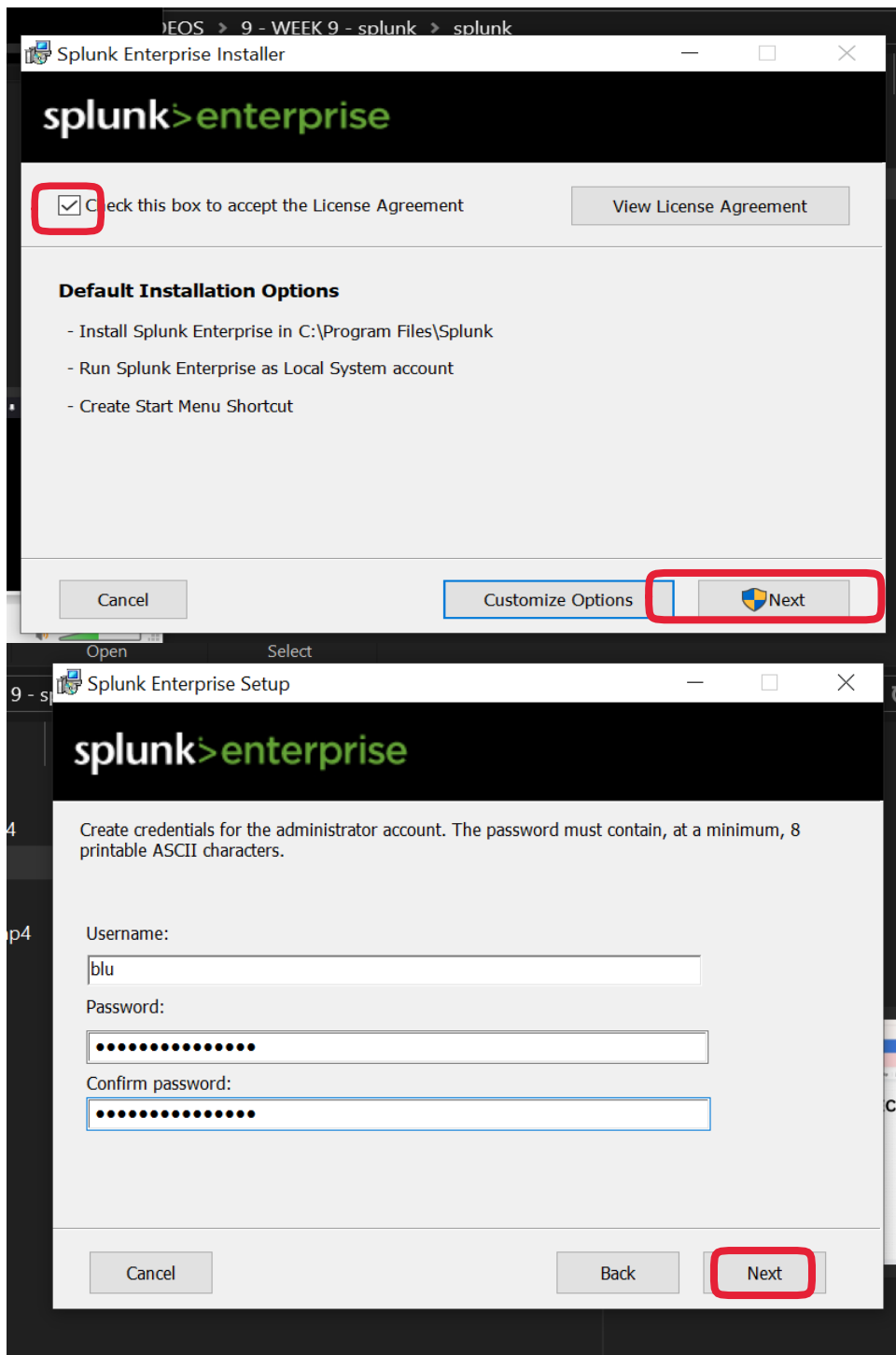
- Host machine
- Windows Server VM (already set up)

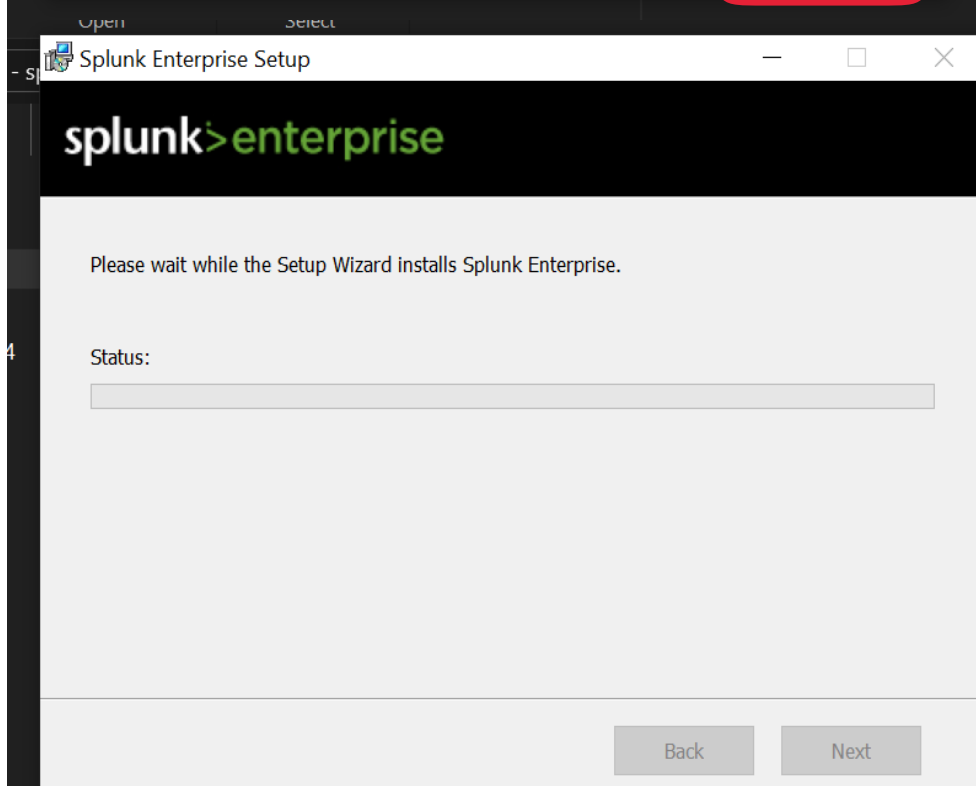
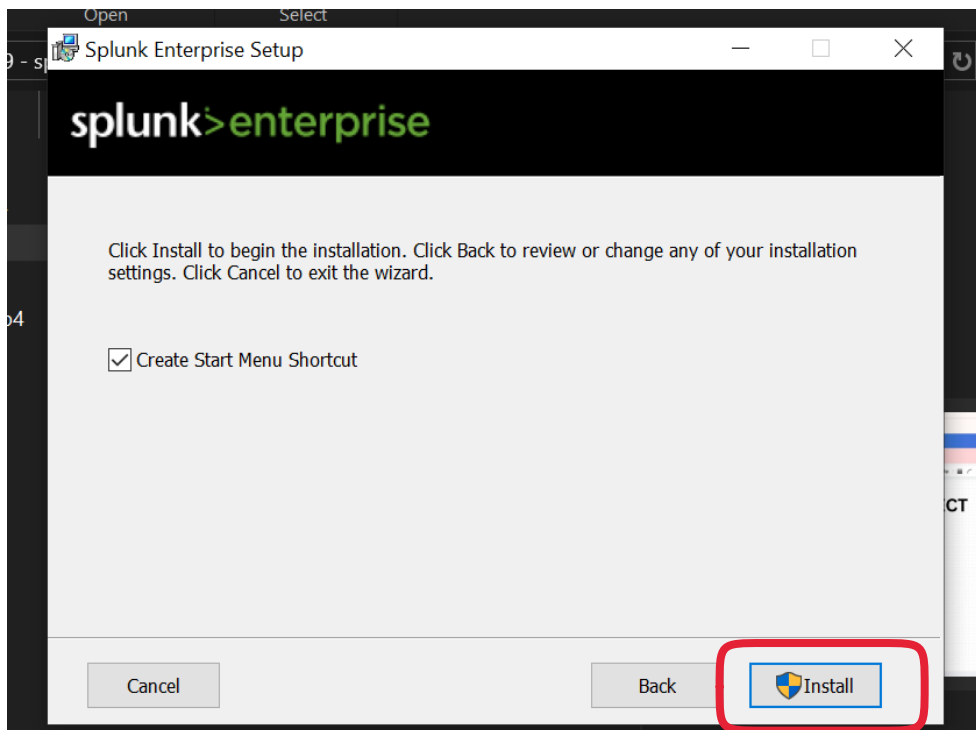
2. Installing Splunk Enterprise

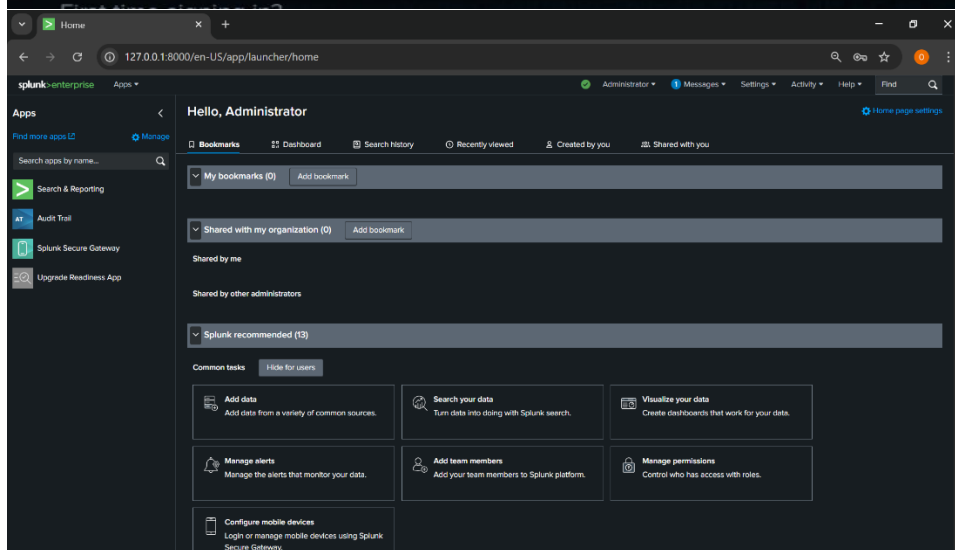
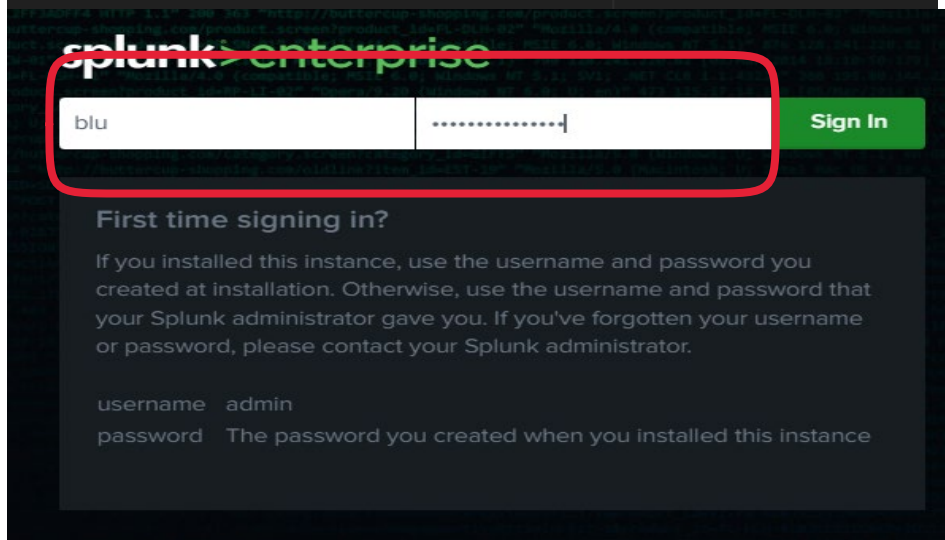
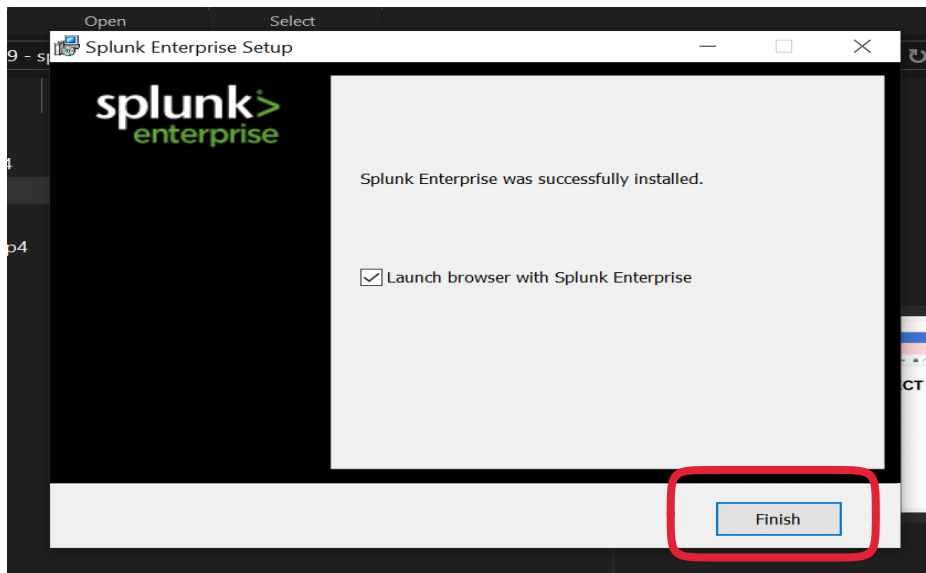
Steps:

Download **Splunk Enterprise**





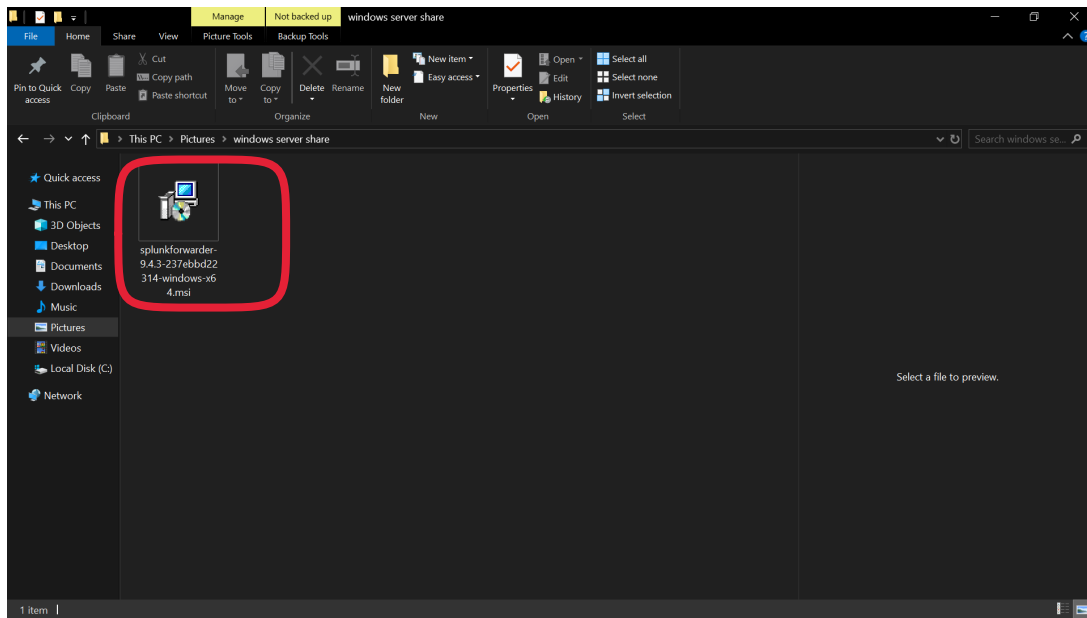


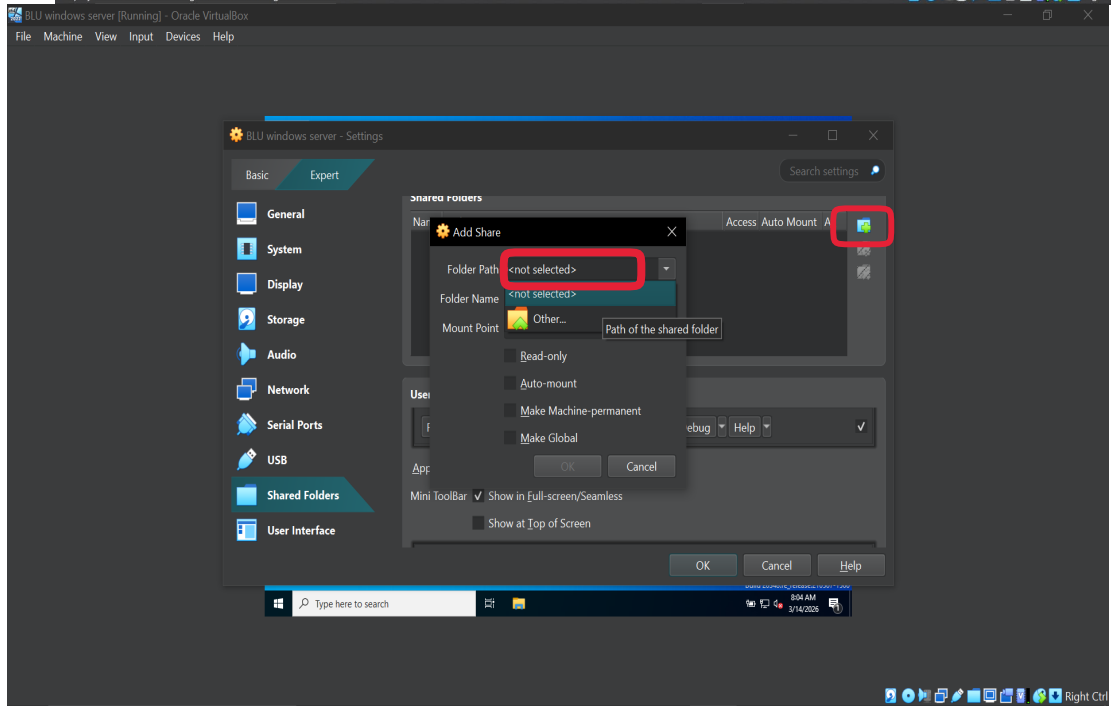
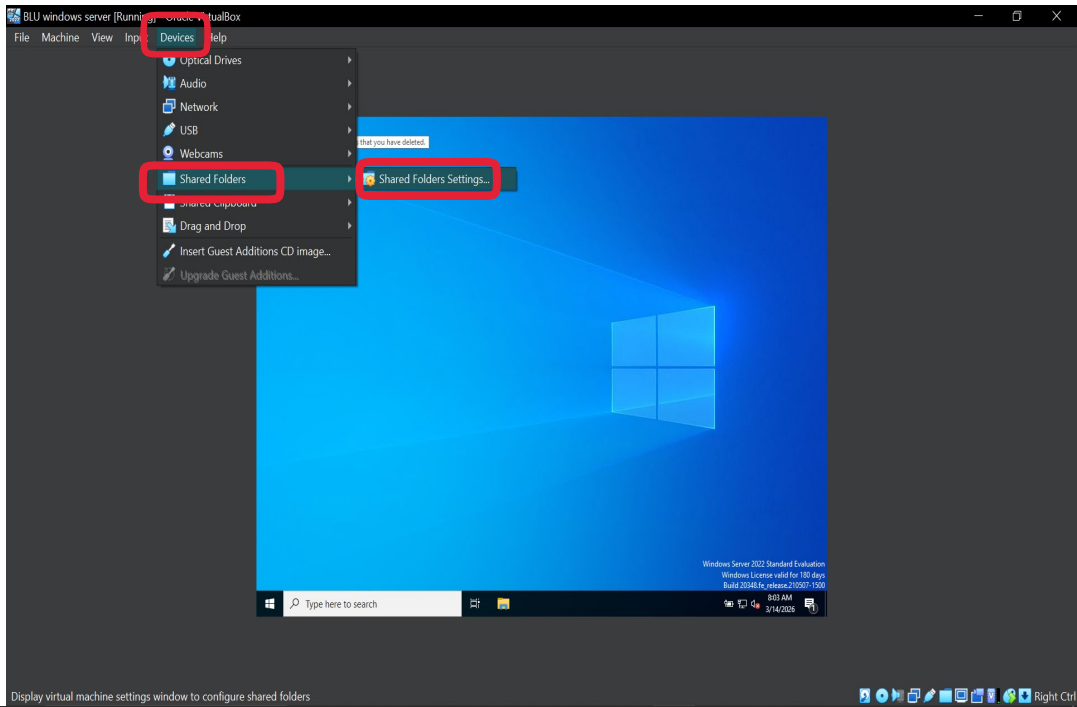


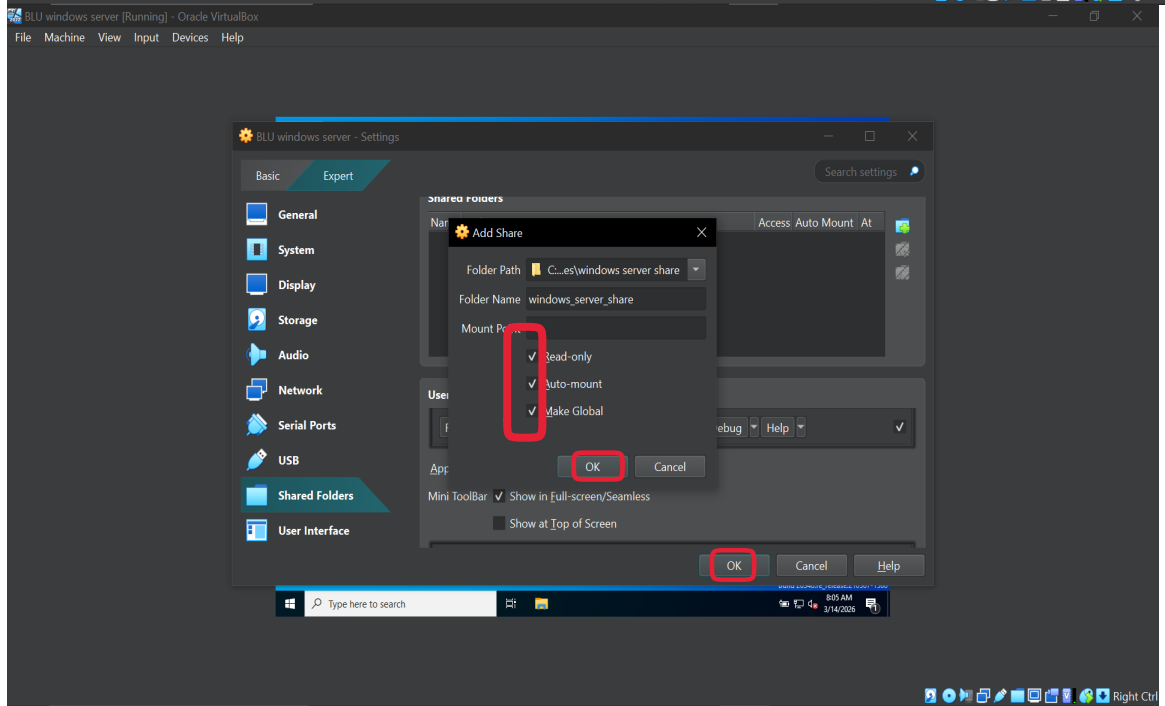
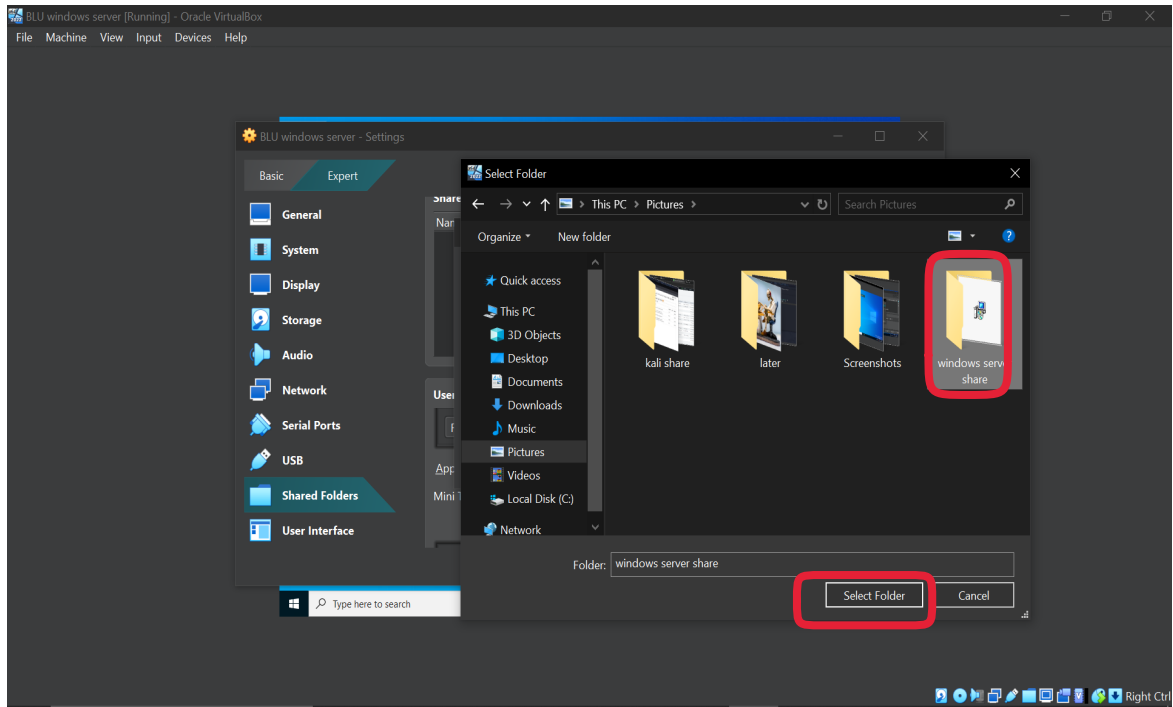
here is the enterprise dashboard

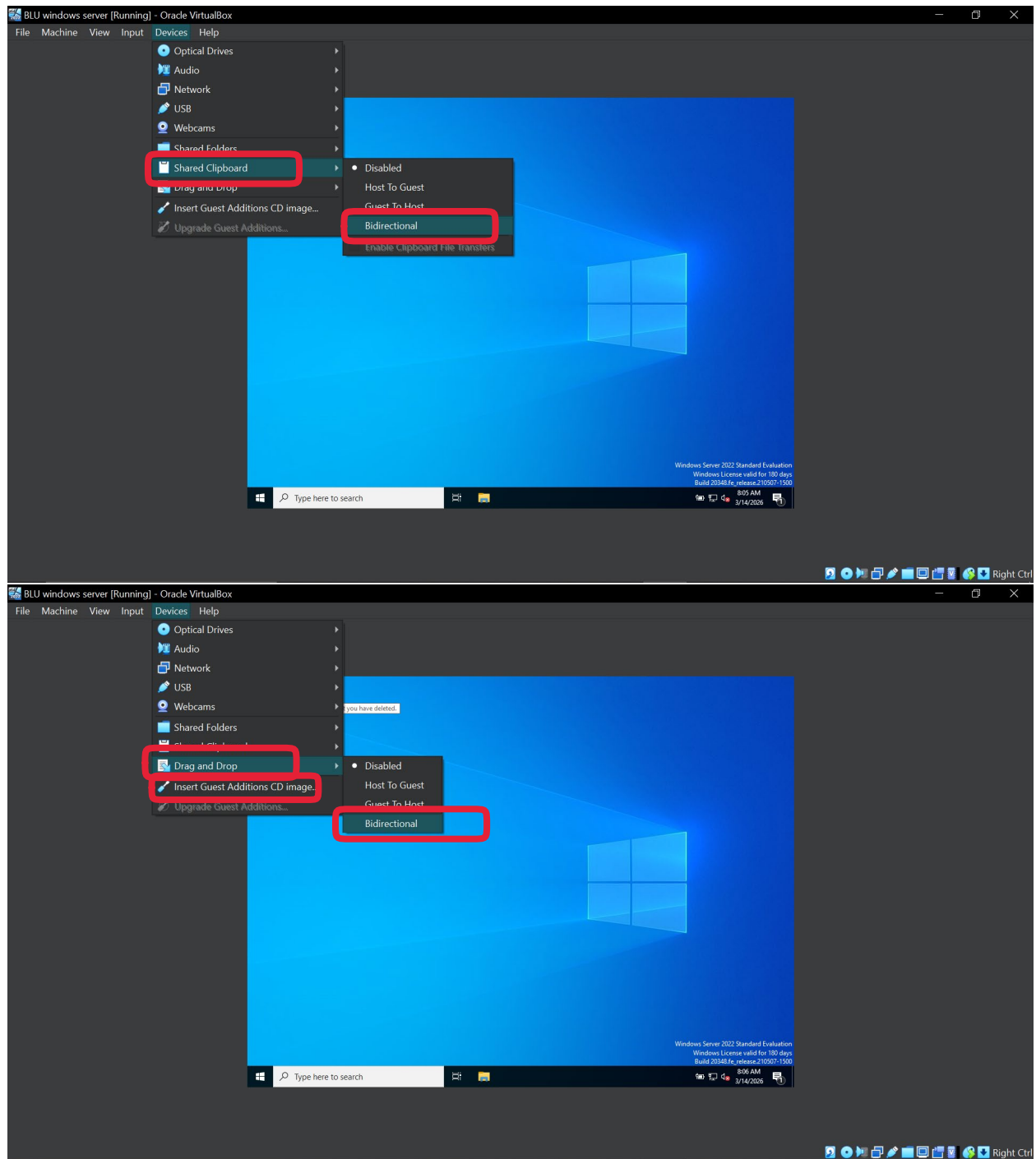
3. Installing Splunk Universal Forwarder

Download the Splunk Universal Forwarder on the host machine and place the installer in a shared folder accessible by the Windows Server VM. Ensure that the Windows Server virtual machine is already running and ready for the installation process.

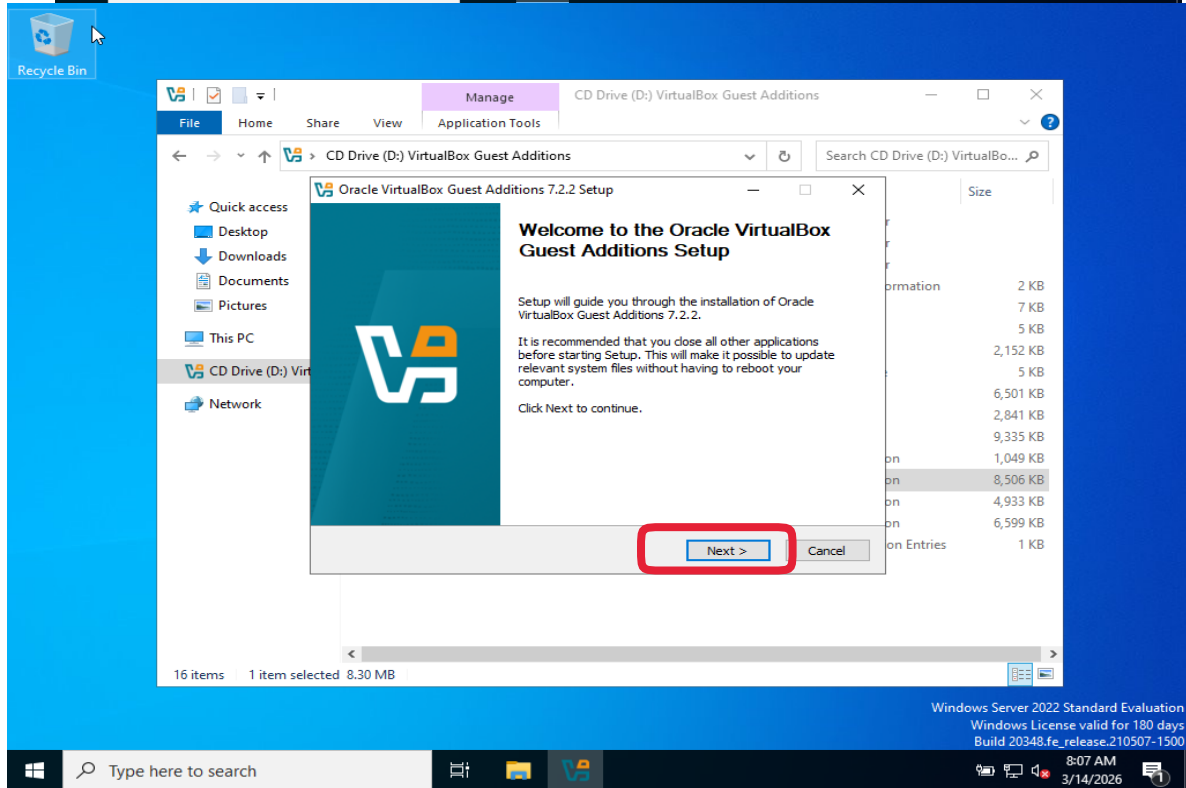
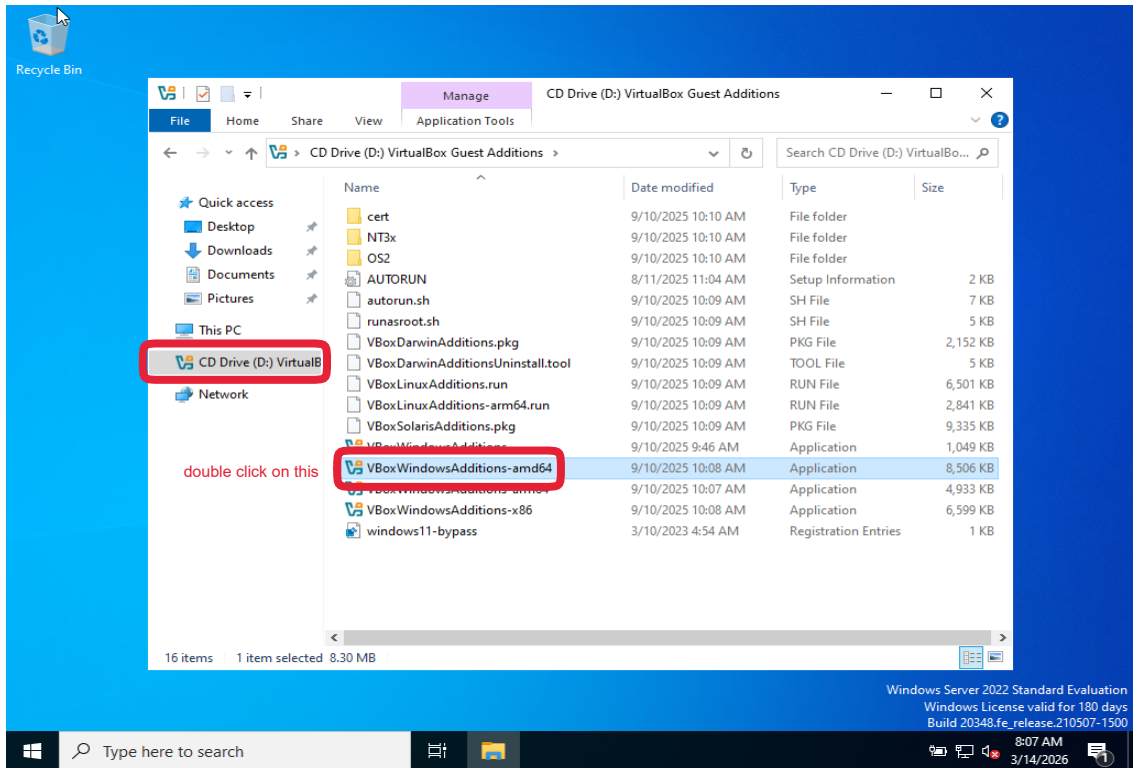


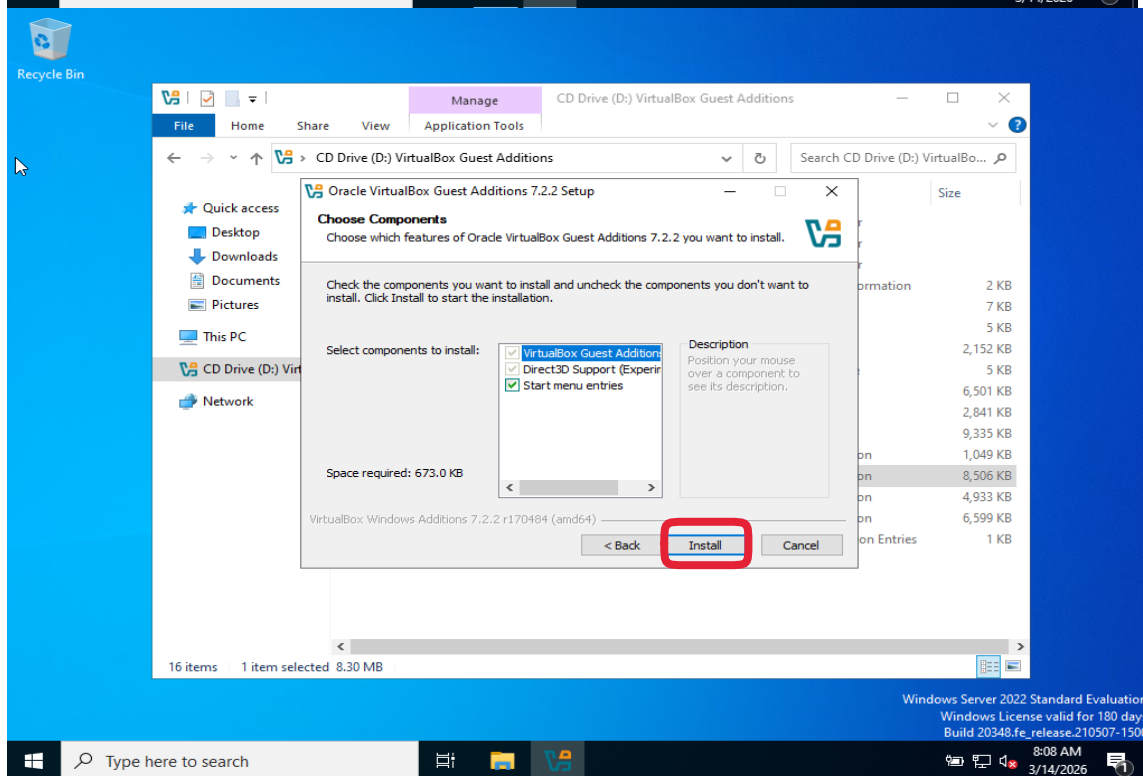
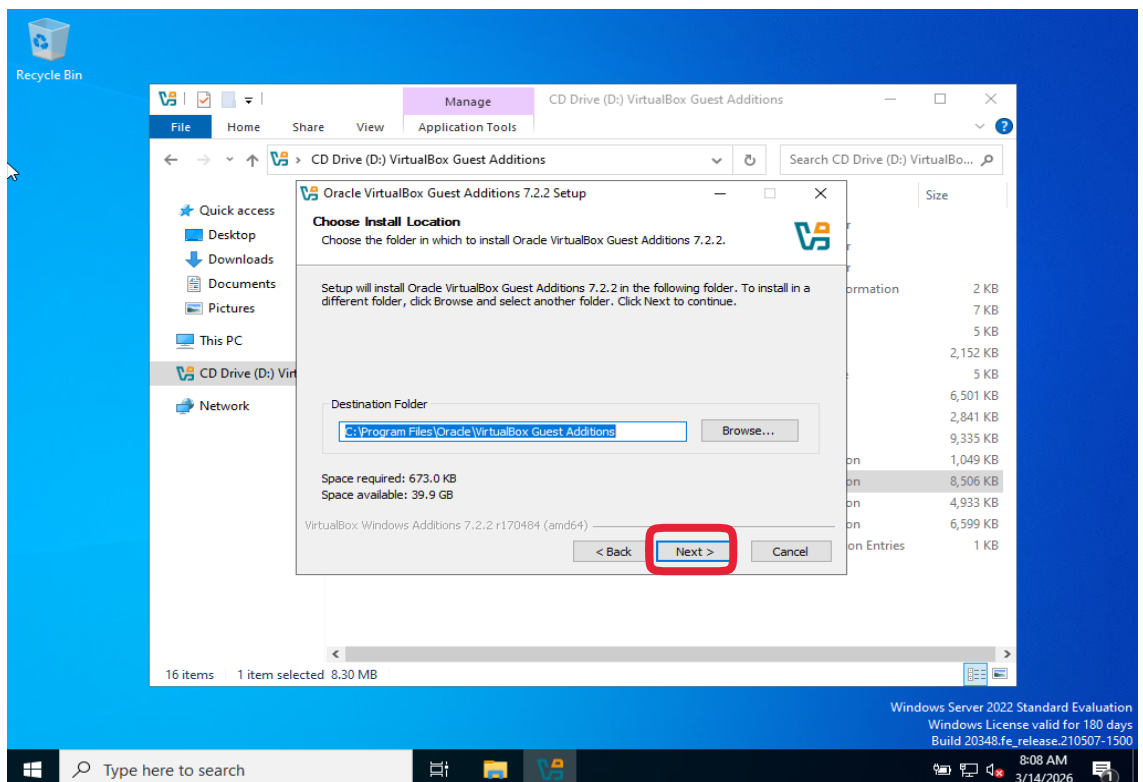


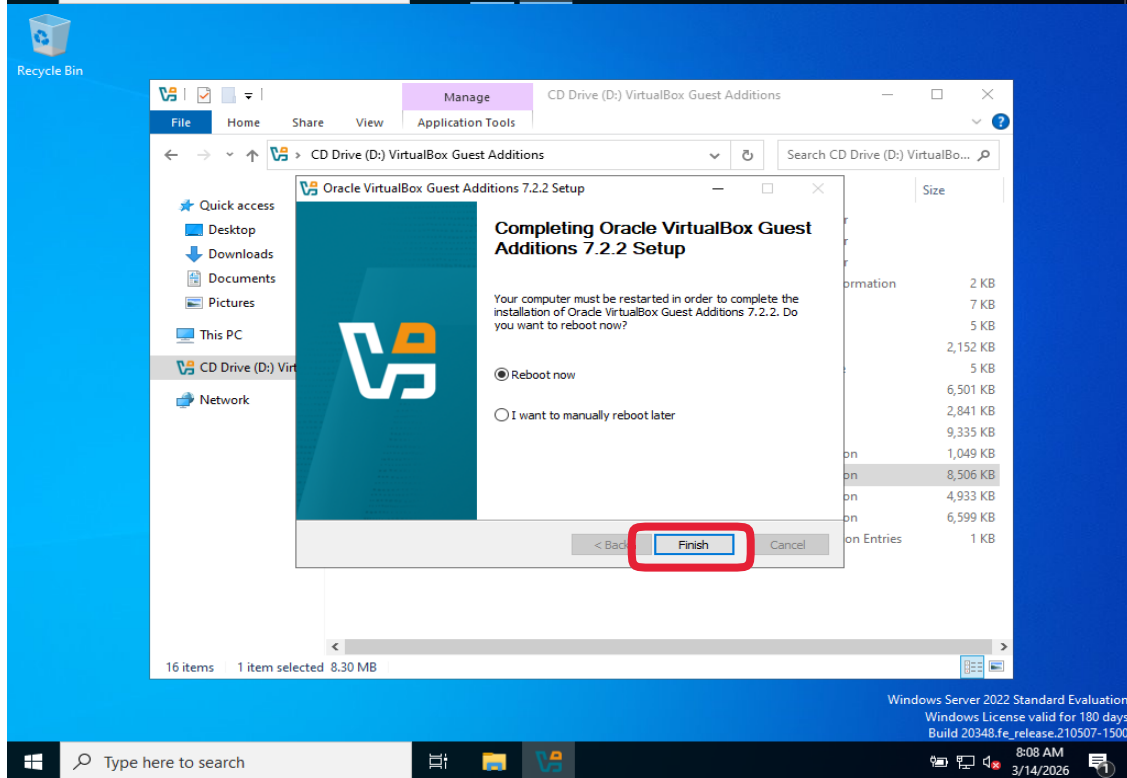
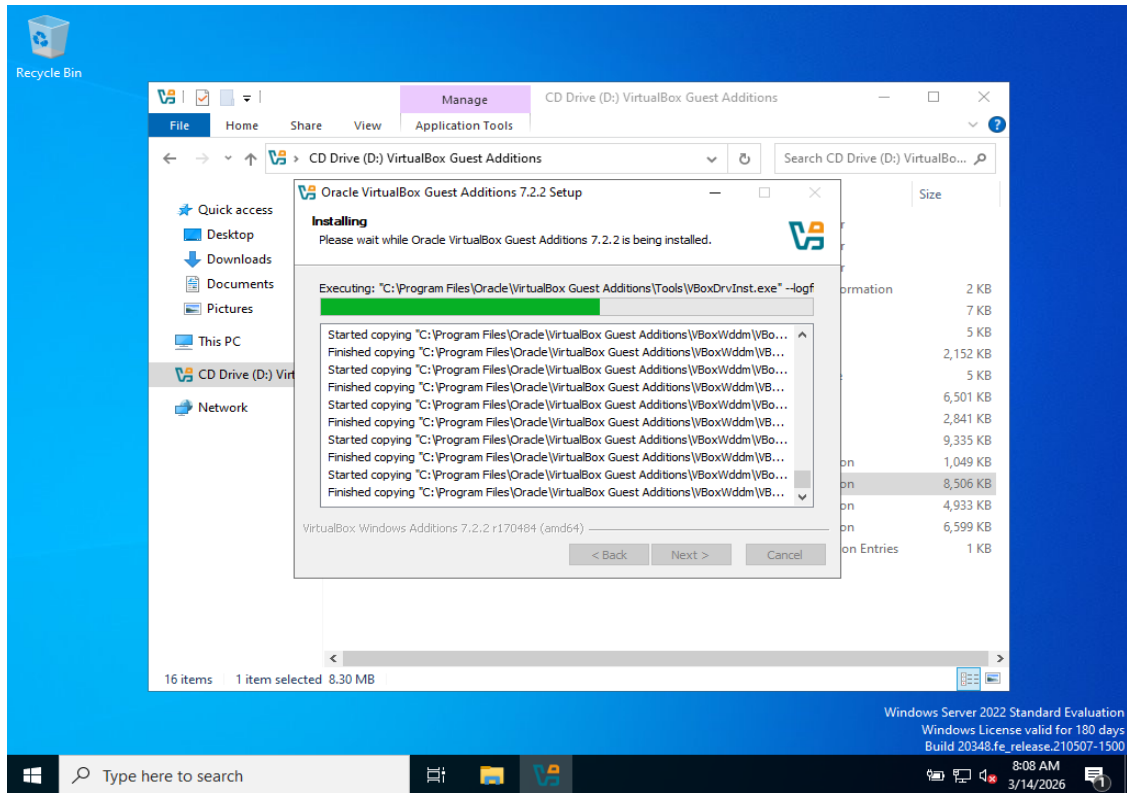


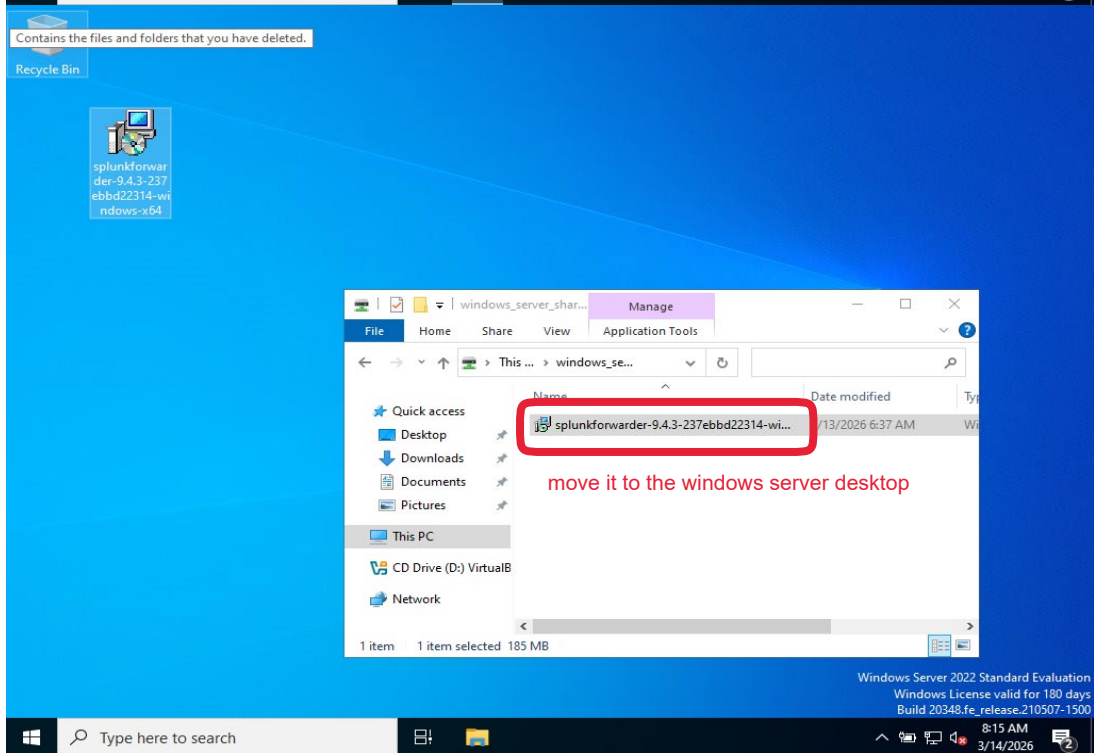
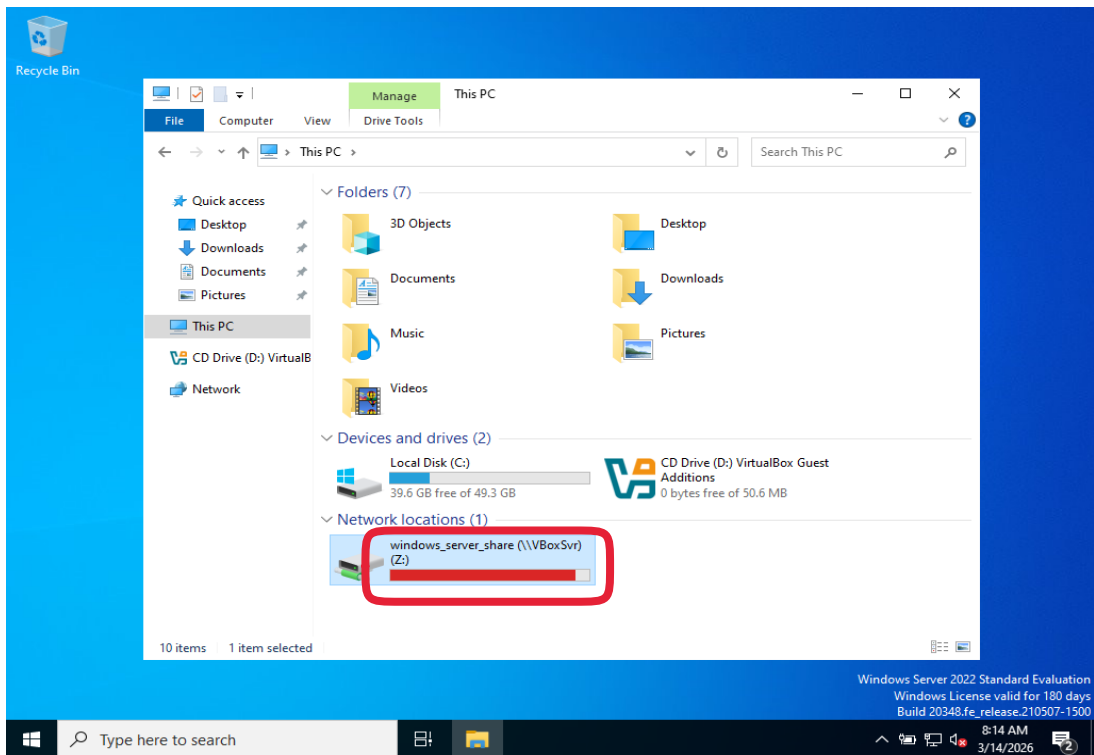


make sure to insert the guest additions CD Image

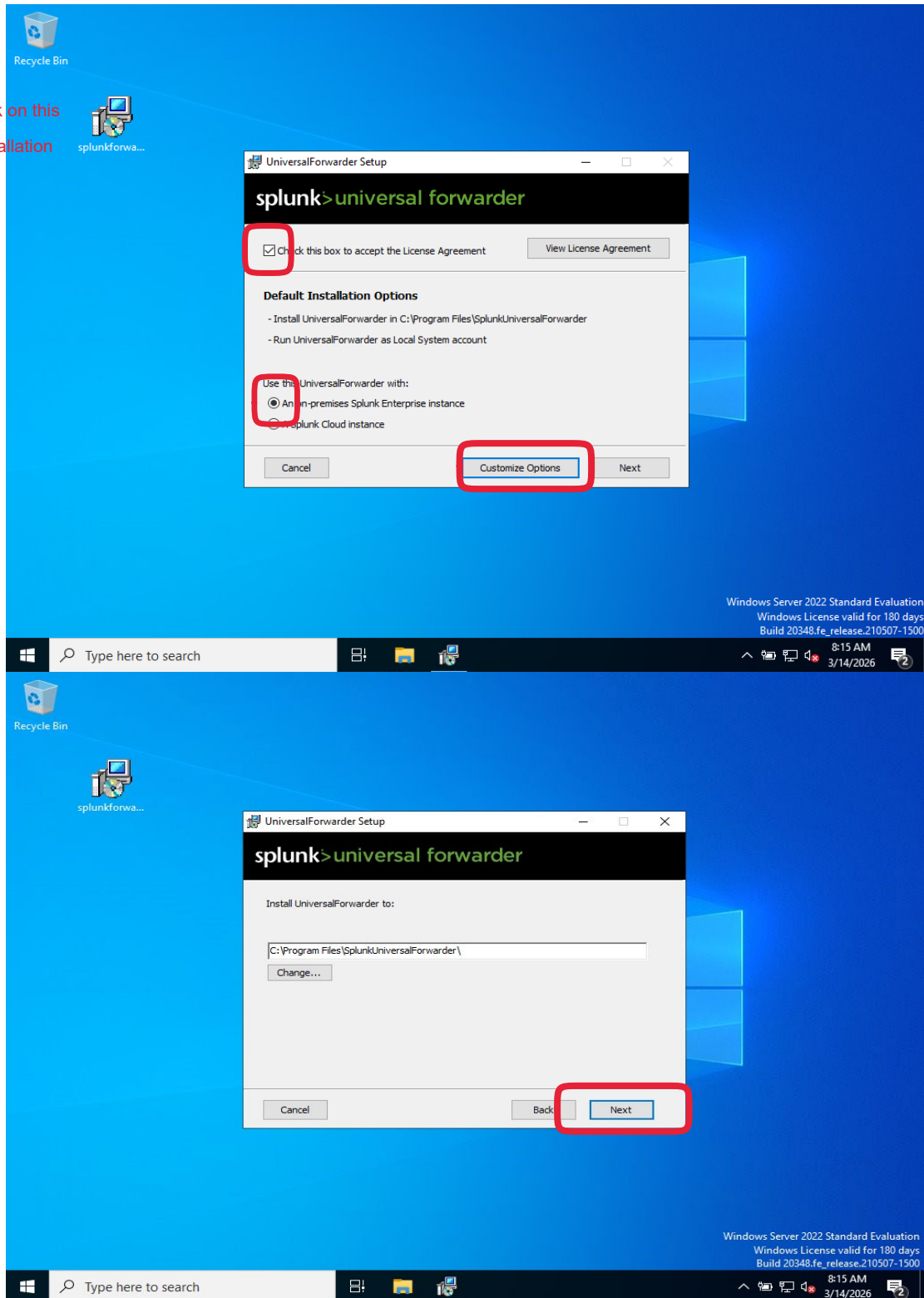


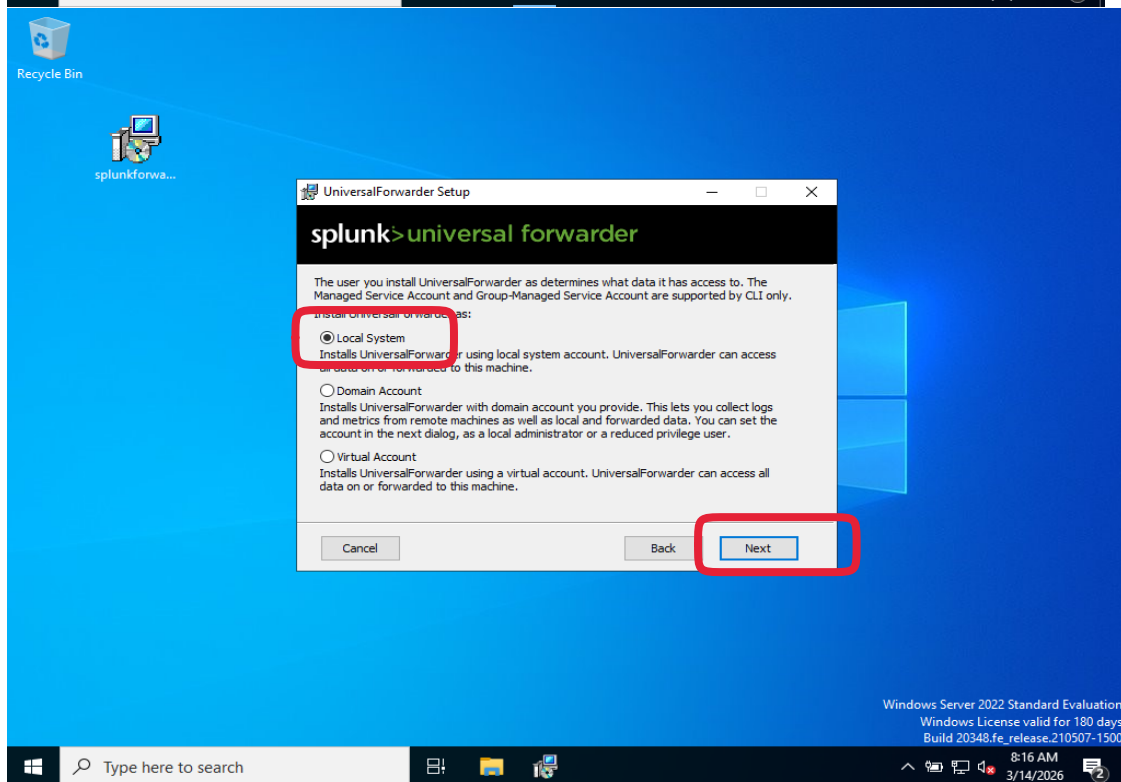
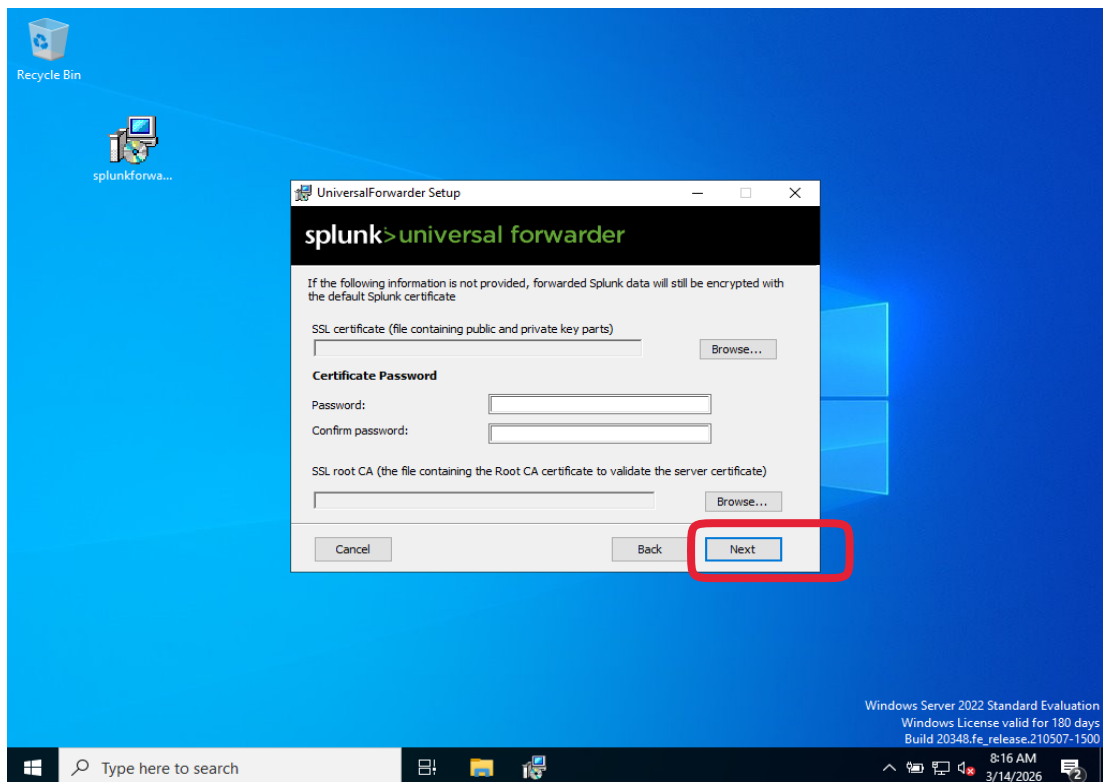


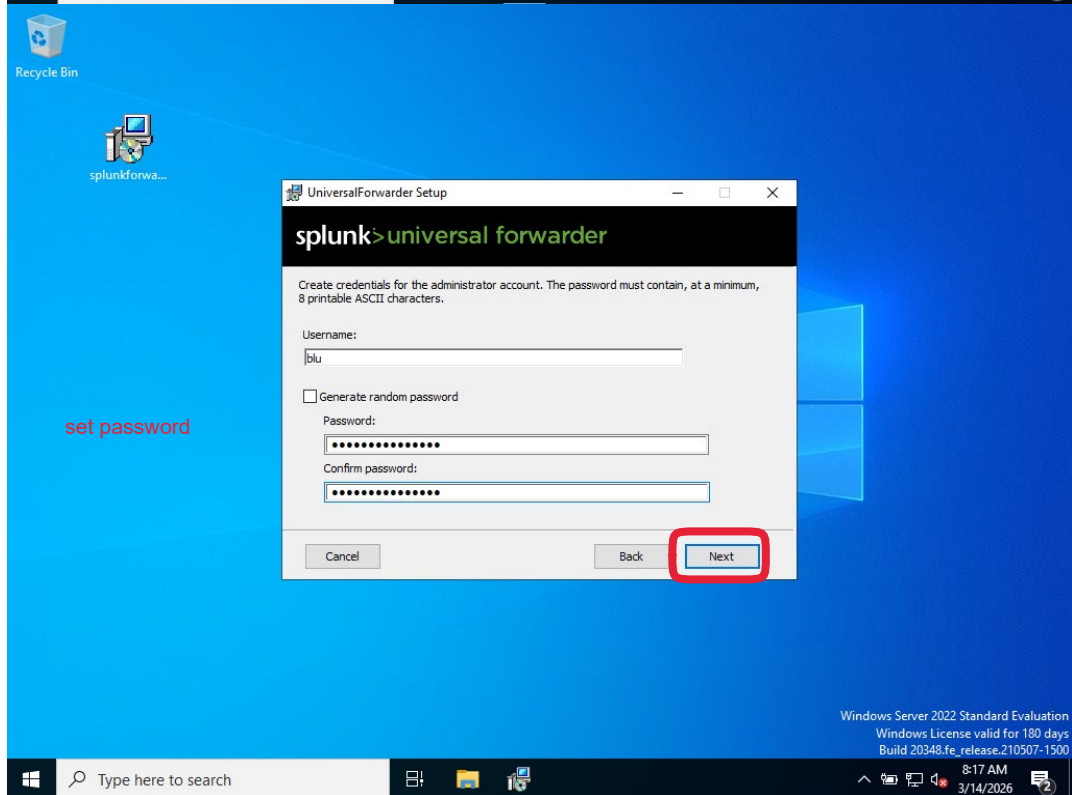
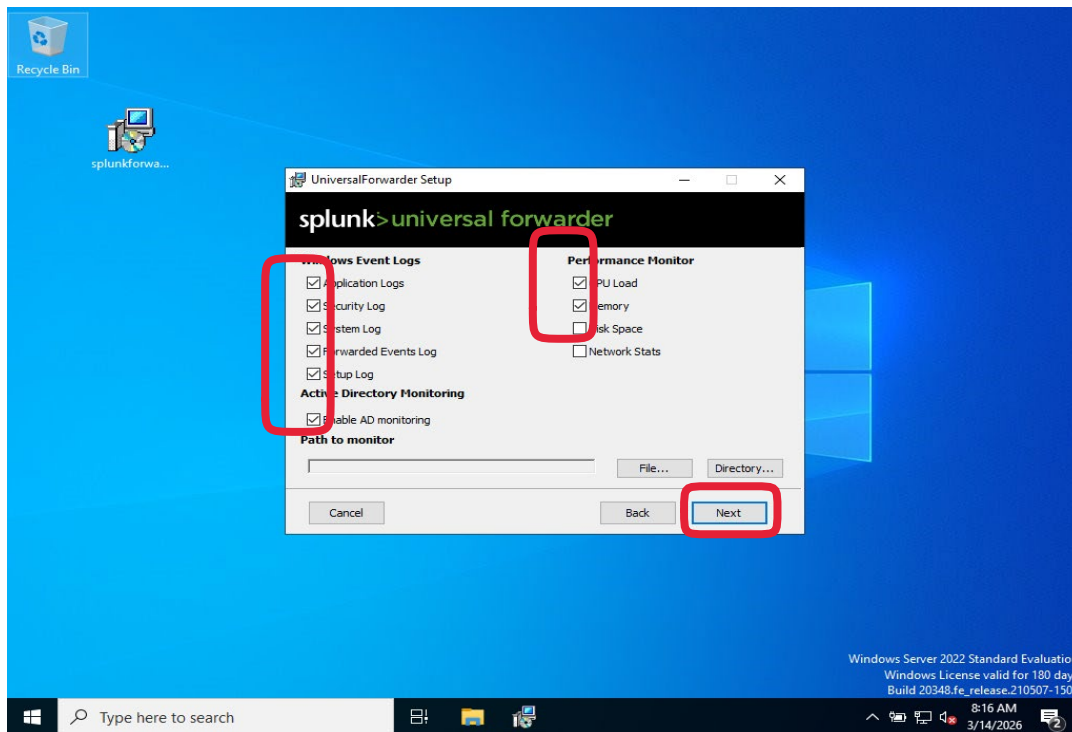


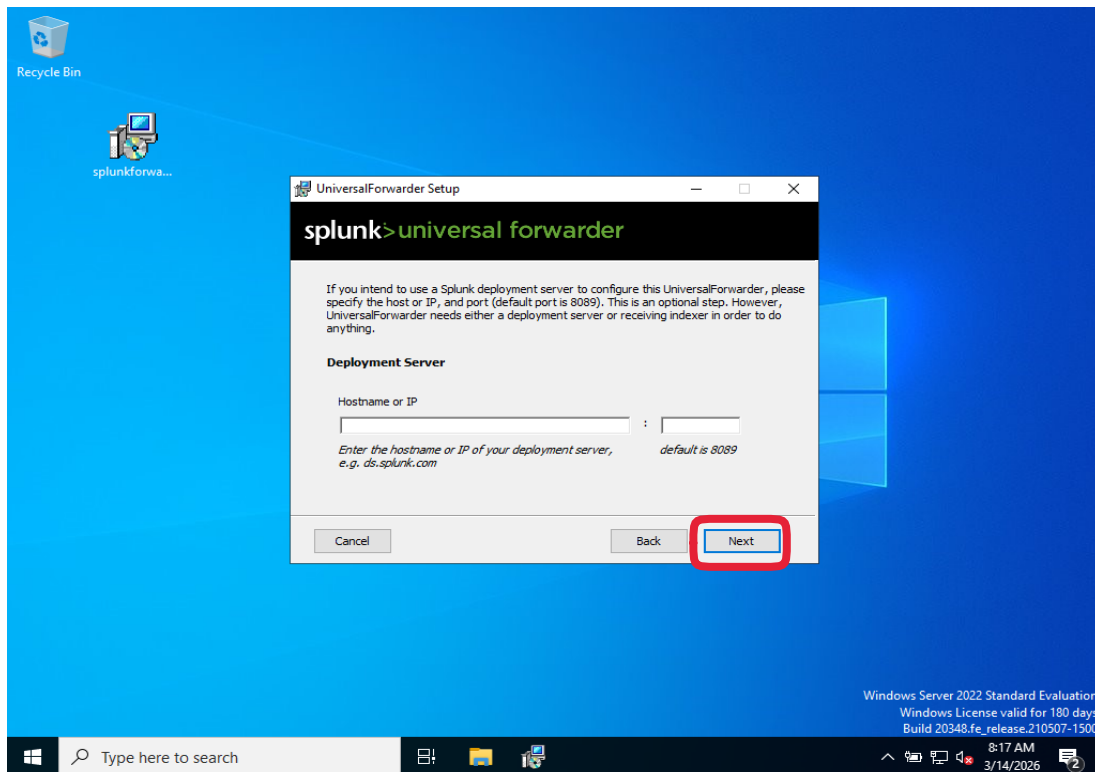


double click on this
to start installation

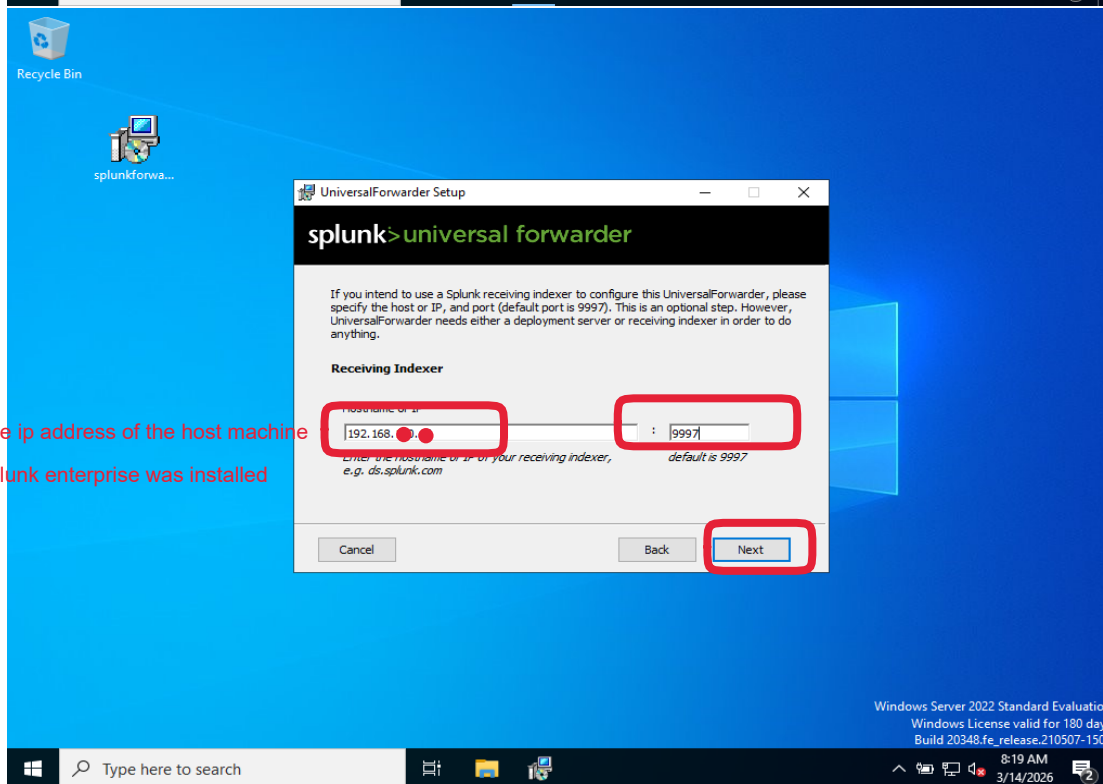


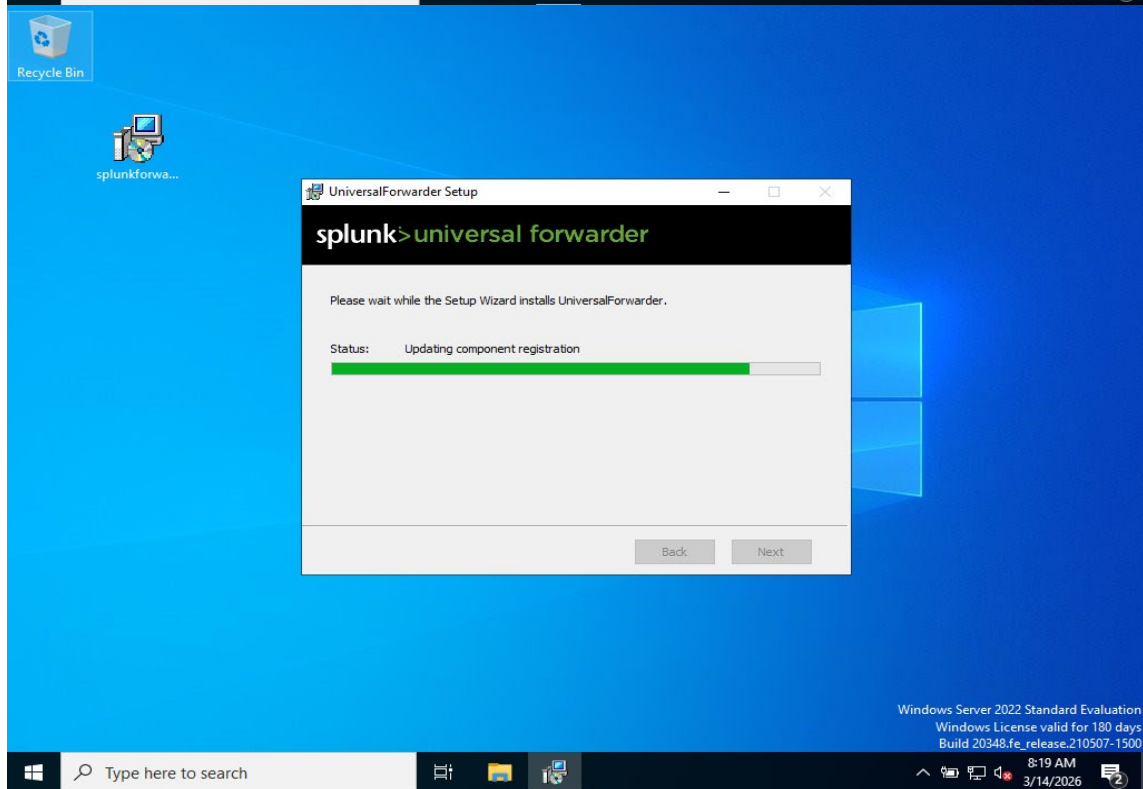
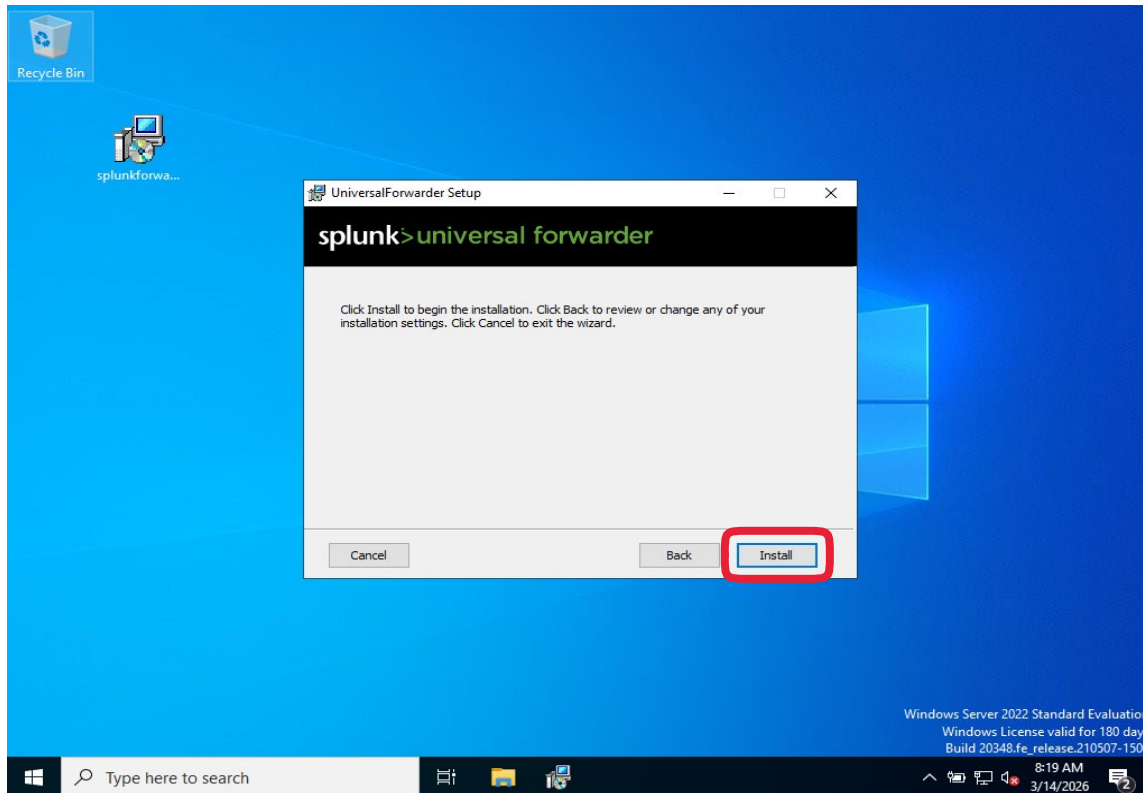


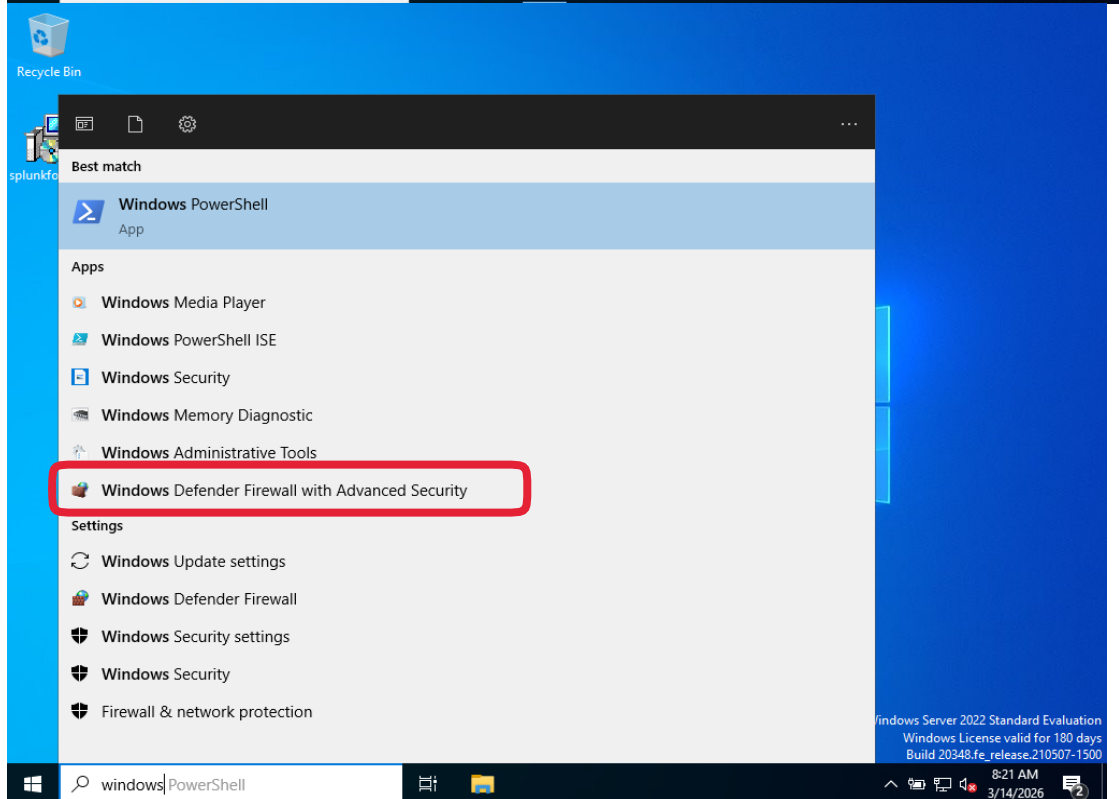
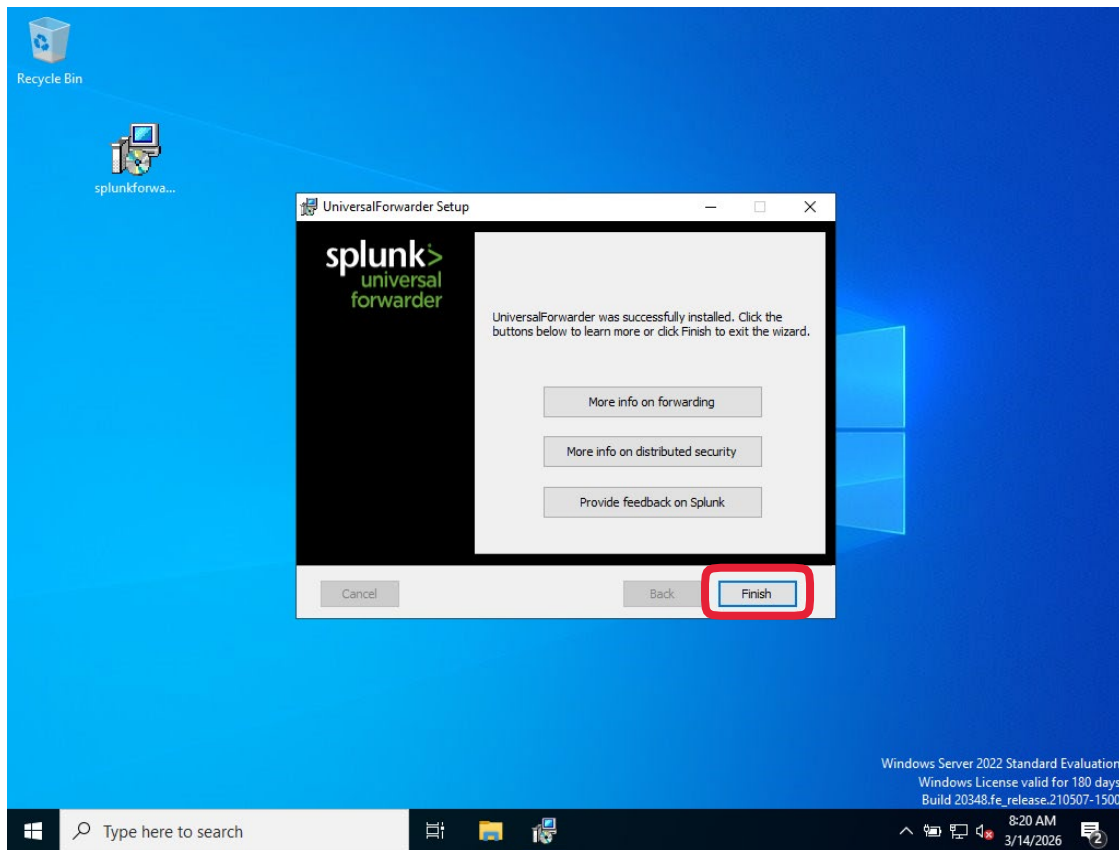




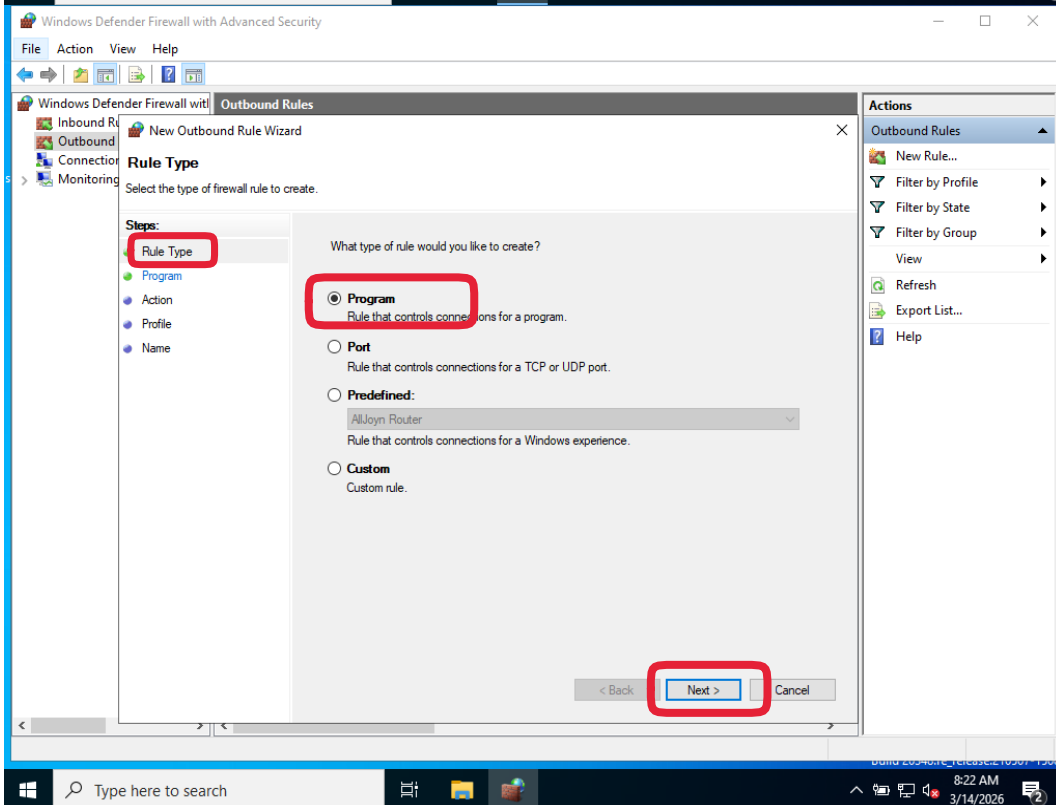
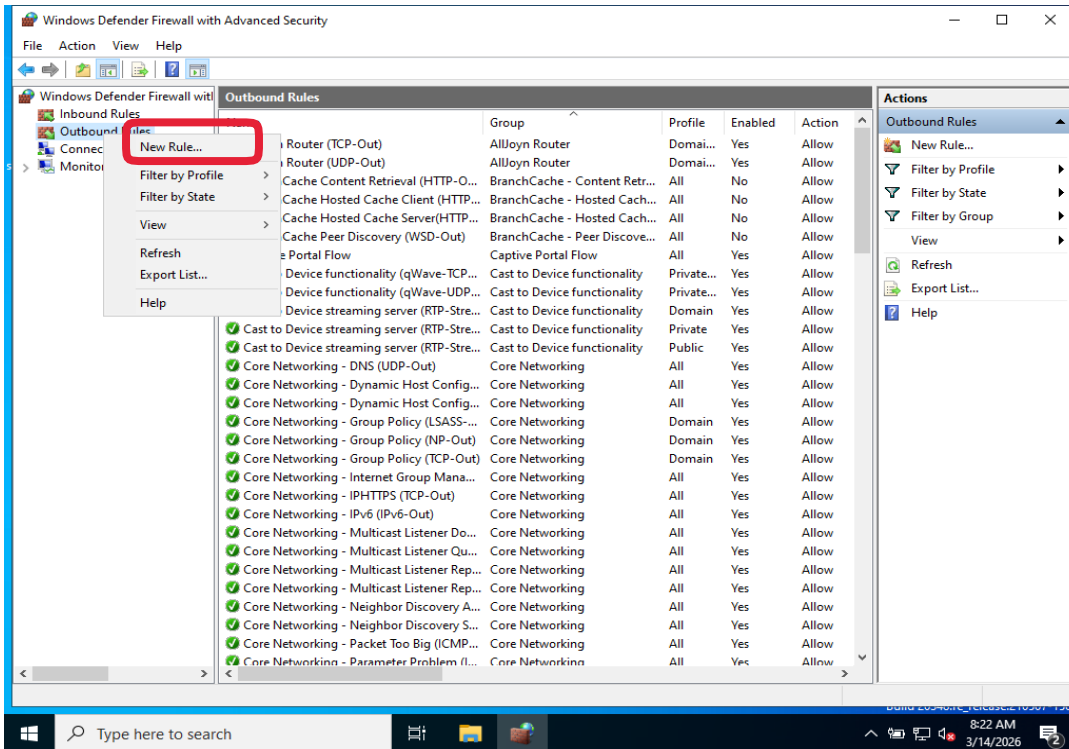
this will be the ip address of the host machine
where the splunk enterprise was installed

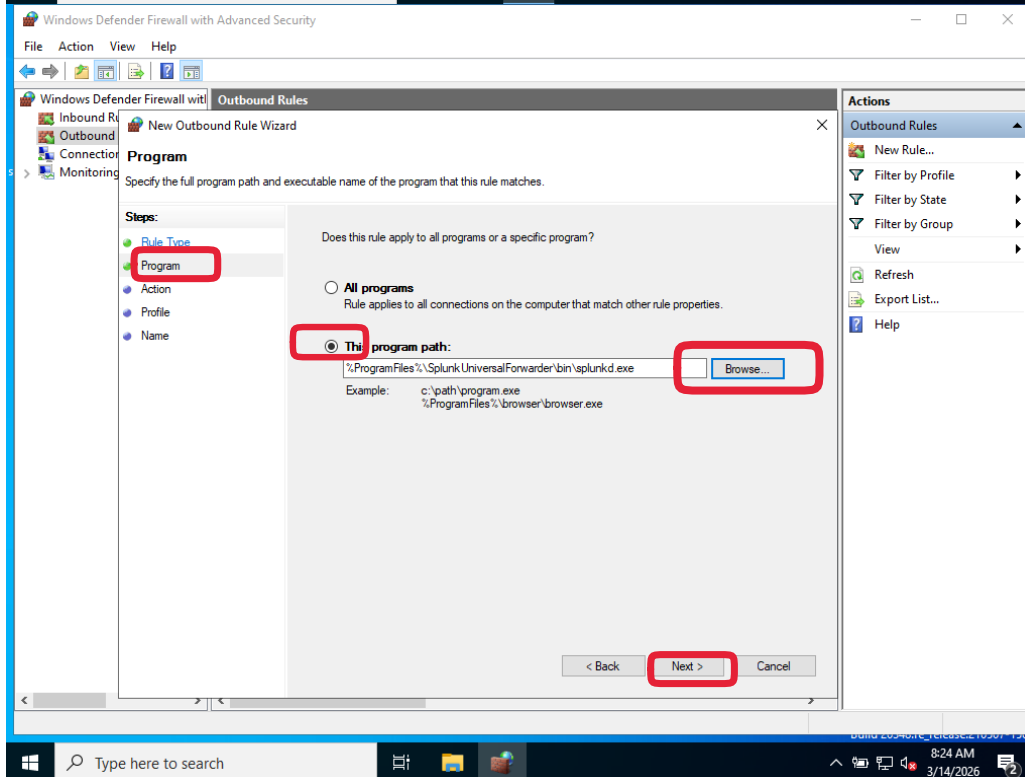
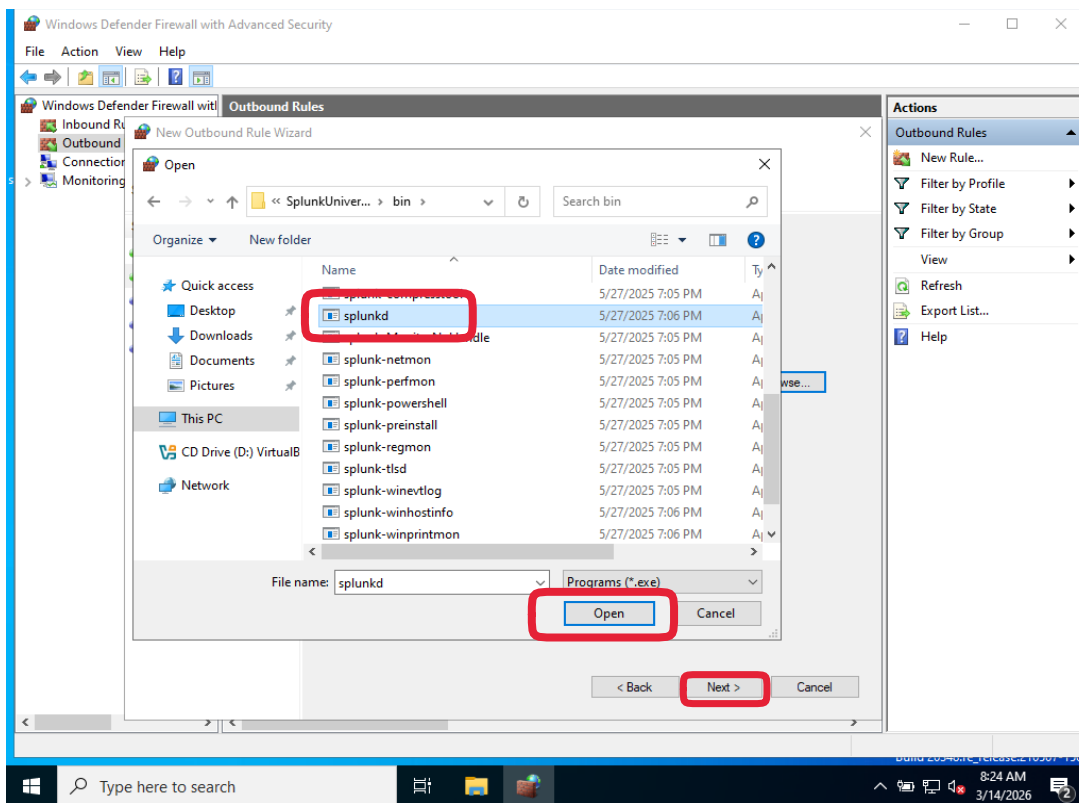


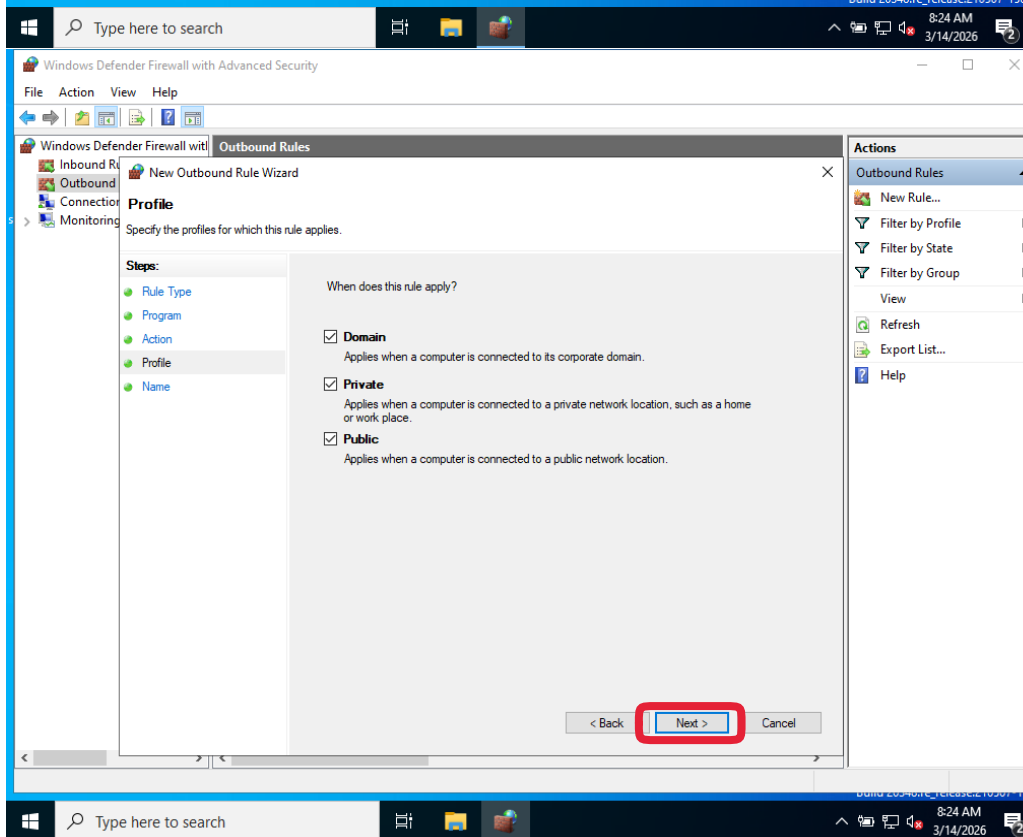
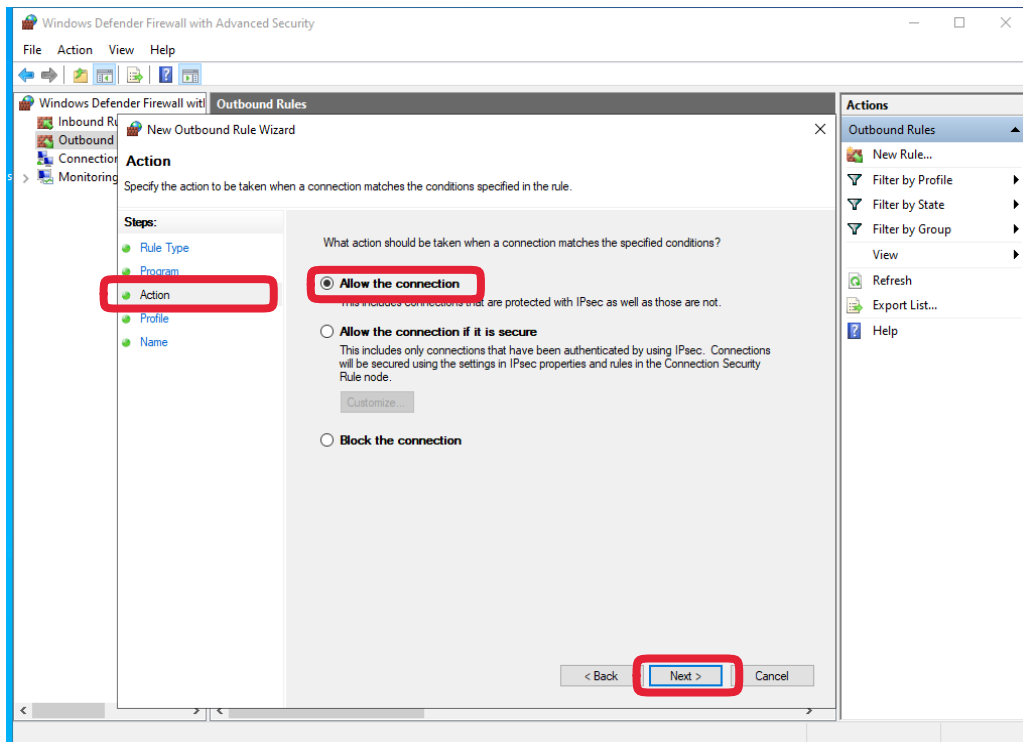


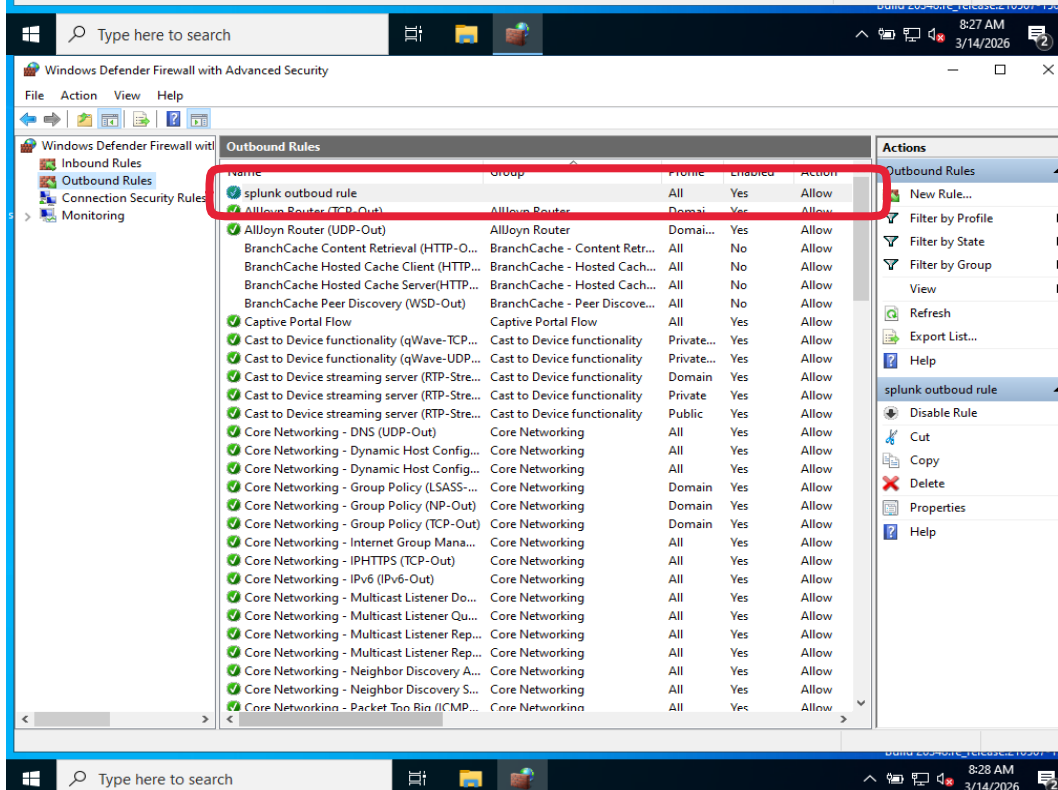
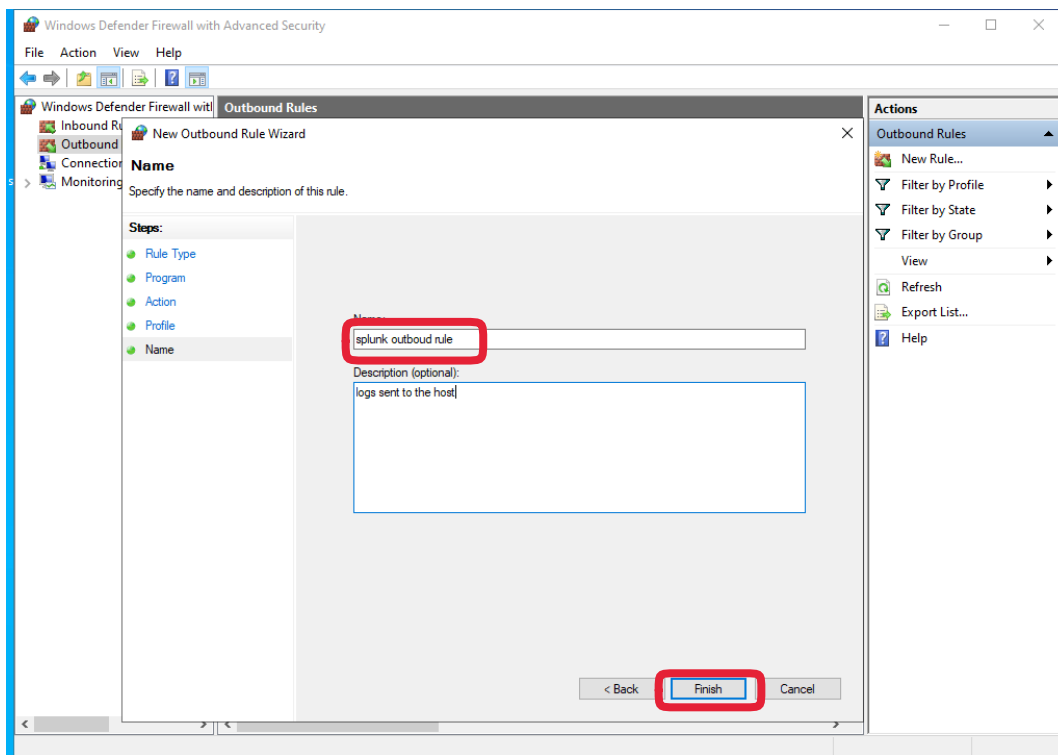


setup an
outbound rule for
the outgoing
traffic

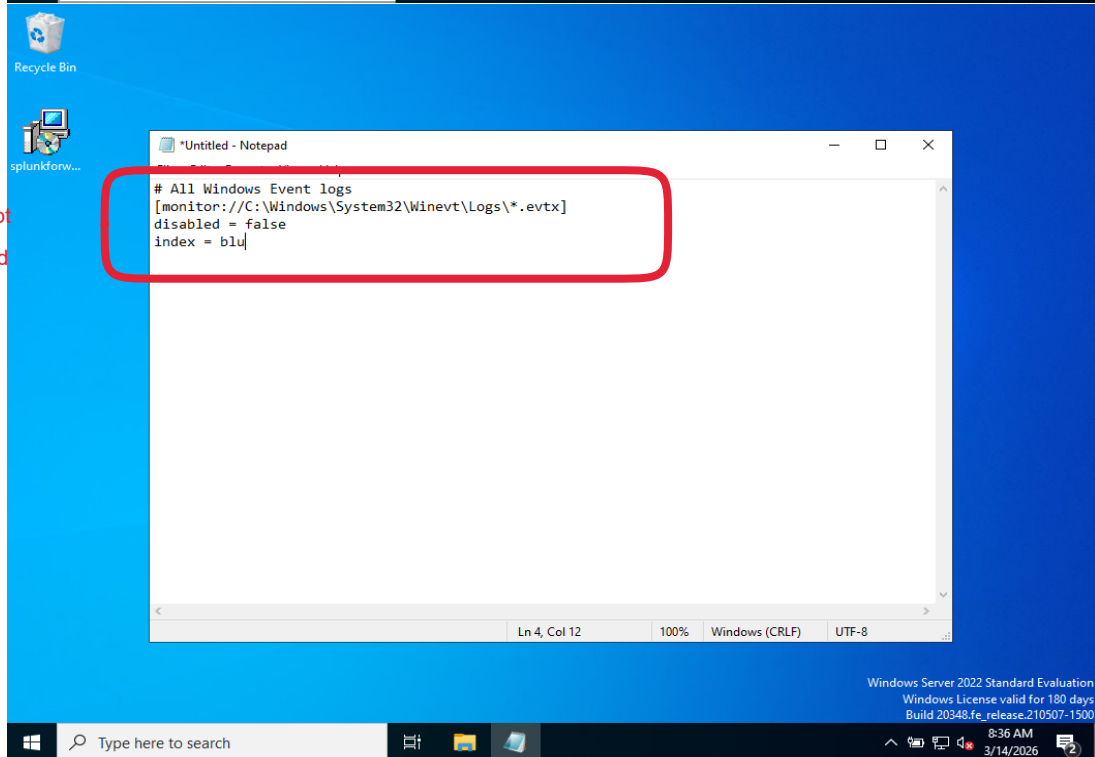
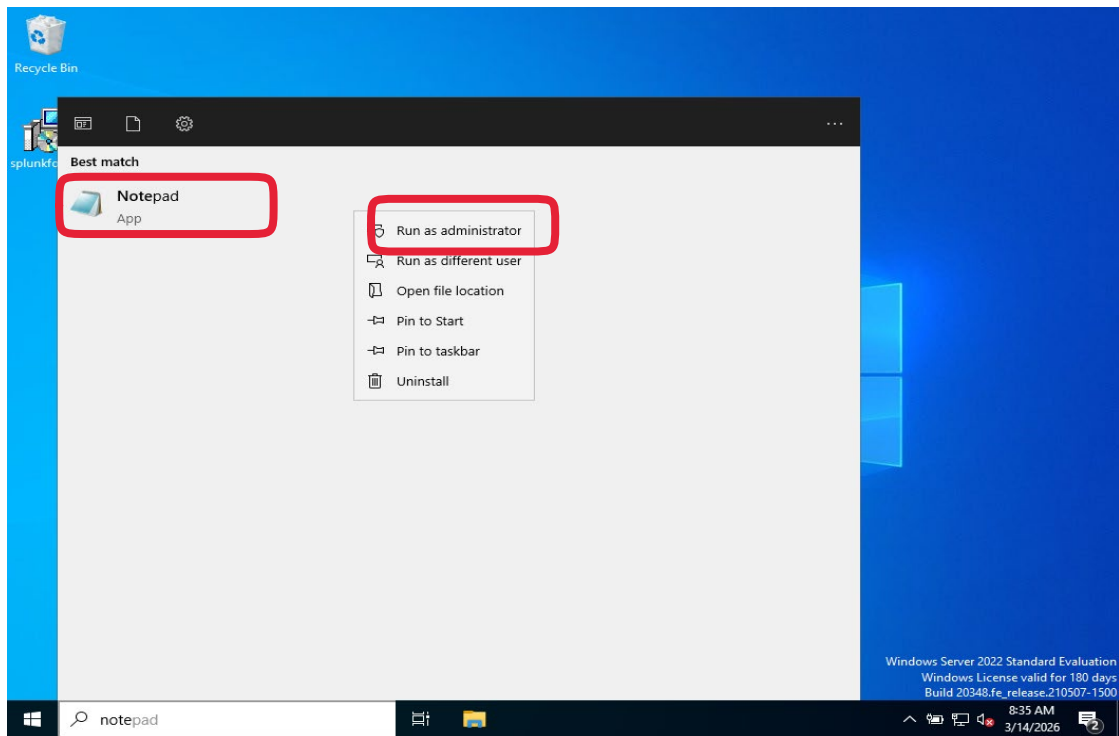




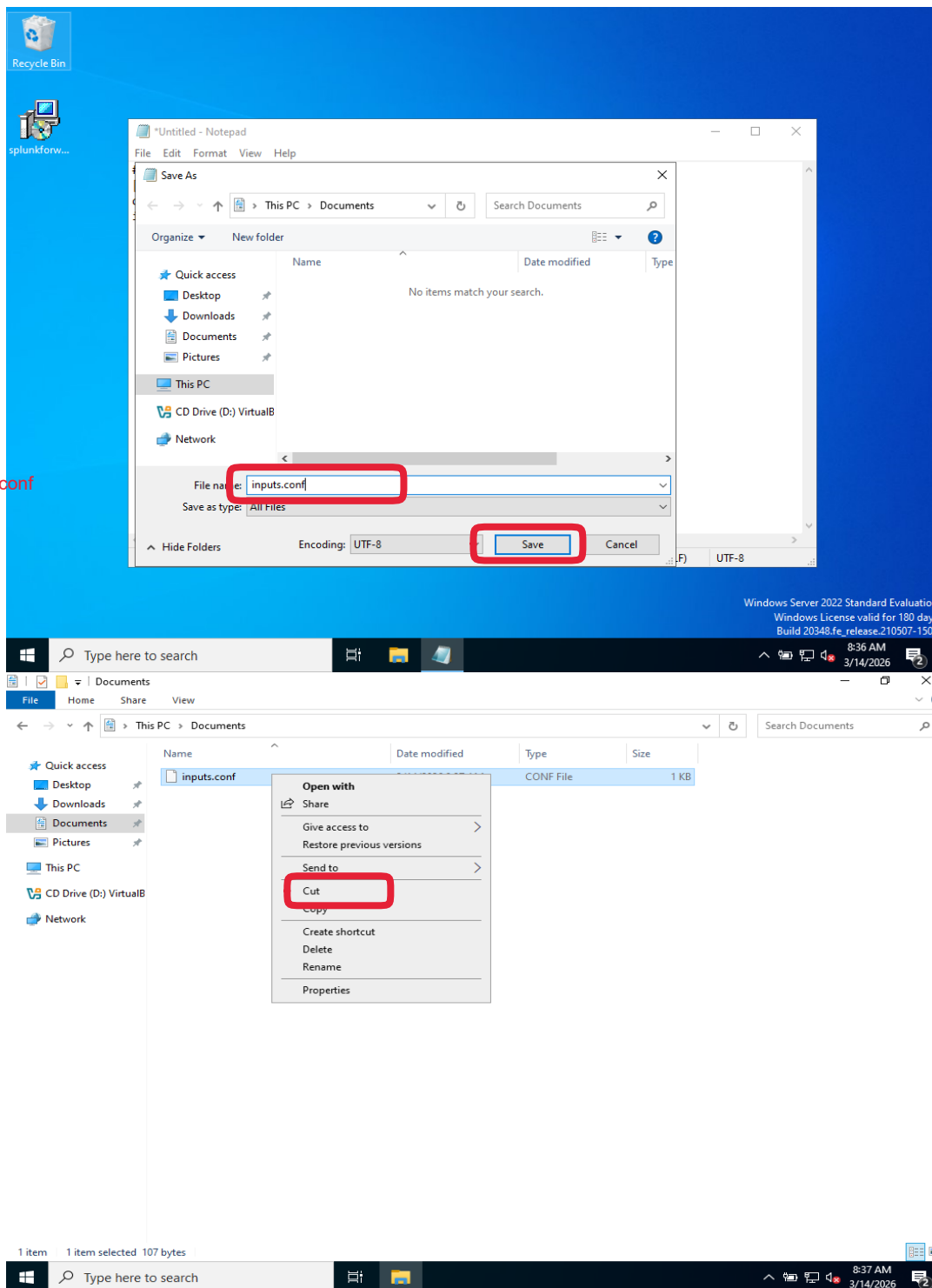




create a log file where the logs are to be stored

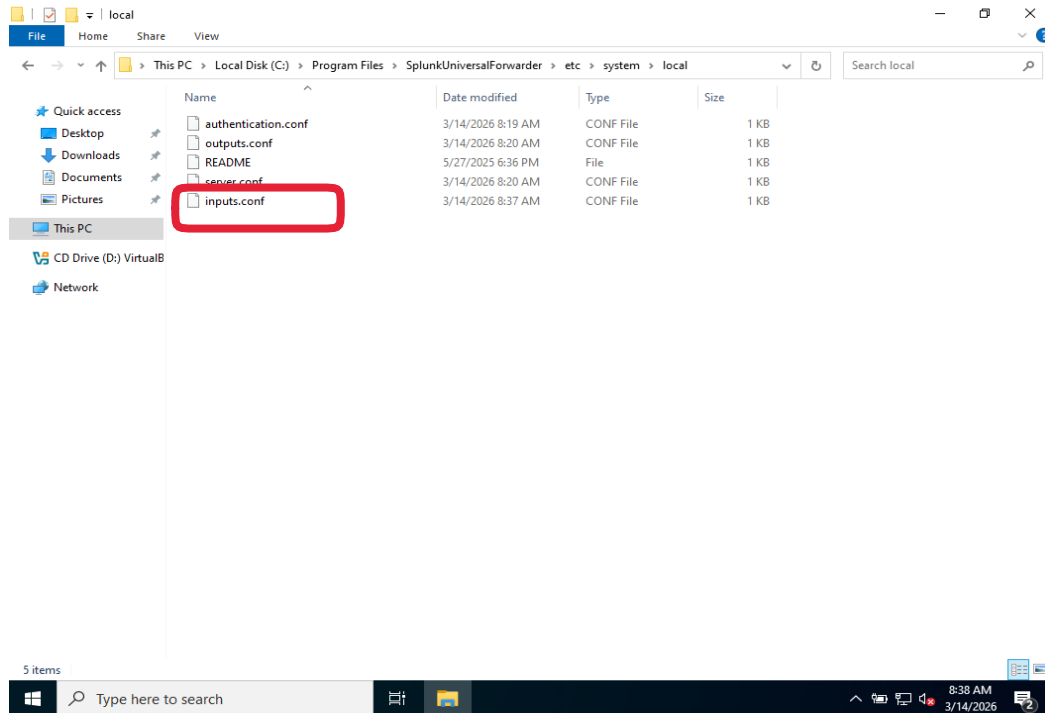


input these script
into the notepad



inputs.conf file should be saved in this directory

C:\Program Files\SplunkUniversalForwarder\etc\system\local\



4. Configuring Log Forwarding

inputs.conf

this was used;

```
[monitor://C:\Windows\System32\Winevt\Logs\*.evtx]
disabled = false
index = blu
```

You can always modify the index name depending on the username you used while registering for you splunk enterprise

5. Log Verification

Log verification on Splunk Enterprise GUI

This is the hostname of the windows server

```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>hostname
WIN-VVG2M7ME5QU
C:\Users\Administrator>
```

127.0.0.1:8000/en-US/app/search/search?q=search%20index*&sid=1773506942.400&display.page.search.mode=smart&dispatch.sample_ratio=1&worklo...

splunk>enterprise Apps Administrator Messages Settings Activity Help Find Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

New Search Save As Create Table View Close

Index Last 24 hours

4 events (3/13/26 5:00:00.000 PM to 3/14/26 5:49:02.000 PM) No Event Sampling

Events (4) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 hour per column

Format Show: 20 Per Page View List

Time	Event
3/14/26 4:13:37.000 PM	03/14/2026 08:13:37 AM LogName=Application EventCode=1800 EventID=4 ComputerName=WIN-VVG2M7ME5QU host = WIN-VVG2M7ME5QU source = WinEventLog:Application sourcetype = WinEventLog:Application
3/14/26 4:03:56.000 PM	03/14/2026 08:03:56 AM LogName=Application EventCode=1800 EventID=4 ComputerName=WIN-VVG2M7ME5QU host = WIN-VVG2M7ME5QU source = WinEventLog:Application sourcetype = WinEventLog:Application
3/14/26 3:03:56.000 PM	03/14/2026 07:03:56 AM LogName=Application EventCode=1800 EventID=4 ComputerName=WIN-VVG2M7ME5QU host = WIN-VVG2M7ME5QU source = WinEventLog:Application sourcetype = WinEventLog:Application

SELECTED FIELDS
host 2
source 1
sourcetype 1

INTERESTING FIELDS
ComputerName 2
EventCode 1
EventType 1
Index 1
Keywords 1
Linecount 1
LogName 1
Message 1
OpCode 1
Parent 1
RecordNumber 4
Sid 1
SidType 1
SourceName 1

The hostname indicates that the windows server logs are being received

127.0.0.1:8000/en-US/app/search/search?q=search%20index*&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=&earliest=-2...

Events (4) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect 1 hour per column

Format Show: 20 Per Page View List

Hide Fields All Fields

SELECTED FIELDS

- host 2
- source 1
- sourcetype 1

INTERESTING FIELDS

- ComputerName 2
- EventCode 1
- EventType 1
- index 1
- Keywords 1
- linecount 1
- LogName 1
- Message 1
- OpCode 1
- punct 1
- RecordNumber 4
- Sid 1
- SidType 1
- SourceName 1
- splunk_server 1
- TaskCategory 1
- Type 1
- User 1

+ Extract New Fields

Time	Event
3/14/26 4:13:37:000 PM	03/14/2026 08:13:37 AM LogName=Application EventCode=1000 EventType=4 ComputerName=WIN-VVG2M7ME5QU Show all 15 lines host = WIN-VVG2M7ME5QU source = WinEventLog:Application sourcetype = WinEventLog:Application
3/14/26 4:03:56:000 PM	03/14/2026 08:03:56 AM LogName=Application EventCode=1000 EventType=4 ComputerName=WIN-VVG2M7ME5QU Show all 15 lines host = WIN-VVG2M7ME5QU source = WinEventLog:Application sourcetype = WinEventLog:Application
3/14/26 3:03:56:000 PM	03/14/2026 07:03:56 AM LogName=Application EventCode=1000 EventType=4 ComputerName=WIN-8KHLETG7P31 Show all 15 lines host = WIN-8KHLETG7P31 source = WinEventLog:Application sourcetype = WinEventLog:Application
3/14/26 2:56:17:000 PM	03/14/2026 06:56:17 AM LogName=Application EventCode=1000 EventType=4 ComputerName=WIN-8KHLETG7P31 Show all 15 lines host = WIN-8KHLETG7P31 source = WinEventLog:Application sourcetype = WinEventLog:Application

127.0.0.1:8000/en-US/app/search/search?q=search%20index%20host%3D%20WIN-VVG2M7ME5QU%20source%3D%20WinEventLog%3AApplication%20&display...

spunk® enterprise Apps Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Index host="WIN-VVG2M7ME5QU" source="WinEventLog:Application"

2 events (2/13/26 5:00:00:000 PM to 3/14/26 5:52:37:000 PM) No Event Sampling

Job

Events (2) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect 1 hour per column

Format Show: 20 Per Page View List

Hide Fields All Fields

SELECTED FIELDS

- host 1
- source 1
- sourcetype 1

INTERESTING FIELDS

- ComputerName 1
- EventCode 1
- EventType 1
- index 1
- Keywords 1
- linecount 1
- LogName 1
- Message 1
- OpCode 1
- punct 1
- RecordNumber 2
- Sid 1

Time	Event
3/14/26 4:13:37:000 PM	03/14/2026 08:13:37 AM LogName=Application EventCode=1000 EventType=4 ComputerName=WIN-VVG2M7ME5QU Show all 15 lines host = WIN-VVG2M7ME5QU source = WinEventLog:Application sourcetype = WinEventLog:Application
3/14/26 4:03:56:000 PM	03/14/2026 08:03:56 AM LogName=Application EventCode=1000 EventType=4 ComputerName=WIN-VVG2M7ME5QU Show all 15 lines host = WIN-VVG2M7ME5QU source = WinEventLog:Application sourcetype = WinEventLog:Application

127.0.0.1:8000/en-US/app/search/search?q=search index%20host%3D%20WIN-VVG2M7ME5QU%20source%3D%20WinEventLog%3AApplication%20&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=&earliest=-24h%40...

Events captured